

MACHINE LEARNING PROJECT

IFT712

COURSE PROJECT

Leaf Classification

*A comparison of six machine-learning model families
(Kaggle Leaf Classification)*

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This personal portfolio edition revises the presentation of the report submitted in April 2026. It preserves the protocol, results, and scientific conclusions of the archived submission; the revisions concern language, layout, and translation.

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1 Introduction

1.1 Context

This project studies Kaggle’s **Leaf Classification** dataset as a tabular multiclass classification problem. Each leaf is described by pre-extracted numerical variables rather than by a raw image, shifting the focus to preprocessing quality, model choice, and, above all, the rigor of the evaluation protocol.

The difficulty stems less from class imbalance than from the ratio between the number of species and the small number of examples available for each one. In this setting, modest performance differences can be artificially amplified unless the exploratory stages and confirmatory estimation are kept strictly separate.

1.2 Study objective

This report compares six families of scikit-learn models within a scientifically defensible framework: perceptron, logistic regression, RBF SVM, random forest, decision tree, and MLP. The aim is not merely to identify the highest score, but to establish which gains remain credible after tuning, accounting for between-fold variability, and secondary verification on a frozen corroboration set.

The report therefore follows a straightforward progression: the EDA motivates preprocessing choices; the model chapters clearly distinguish exploration from confirmation; and the overall discussion compares the **tuned** variants in terms of performance, stability, and practical cost.

1.3 Report structure

Section 2 presents the exploratory data analysis and the methodological framework used throughout the study. Section 3 examines each of the six model families separately. Section 4 provides the overall comparison and scientific discussion, followed by a brief conclusion, appendices, and bibliography.

2 Exploratory data analysis

2.1 Purpose and descriptive scope

The exploratory analysis characterizes the structure of the dataset, verifies its integrity, and identifies the main statistical properties likely to influence preprocessing choices. The findings below remain descriptive: they cover the distribution of observations, feature scales, redundancy among descriptors, and the overall organization of the point cloud, without presuming the behavior of any given classification model.

Two methodological precautions apply at this stage. First, visible structure in correlations or reduced-dimensional projections does not constitute evidence of robust separability. Second, the preprocessing hypotheses suggested by this section should be understood as reasonable experimental directions, not as confirmatory conclusions.

2.2 Dataset profile and integrity

The training dataset contains 990 observations and 194 columns: 192 numerical explanatory variables, an `id` identifier, and a `species` class label. The descriptors are divided into three equally sized families, `margin`, `shape`, and `texture`, each containing 64 variables. This organization matters for interpreting the EDA because it distinguishes within-family redundancy from the weaker dependencies between families.

The basic integrity checks reveal no major anomalies: there are no missing values, no complete duplicate rows, no exact duplicates among the explanatory variables alone, and the `id` identifier is unique for every sample. These properties simplify the subsequent analysis because no structural cleaning is required before modeling.

Indicator	Value
Rows	990
Columns	194
Explanatory variables	192
Classes	99
<code>margin</code> variables	64
<code>shape</code> variables	64
<code>texture</code> variables	64
Missing values	0
Complete duplicates	0
Duplicates among explanatory variables	0
Unique <code>id</code>	Yes

Table 1: Dataset profile

2.3 Class distribution

The target variable has a remarkably regular structure: each of the 99 species is represented by exactly 10 observations. This rules out class imbalance in the usual sense and allows descriptive summaries to be interpreted without the risk of mechanical dominance by a few overrepresented species. This regularity does not, however, make the problem easy. With only 10 leaves per species, each class remains sparsely documented, suggesting non-negligible variability in performance estimates during evaluation.

Metric	Value
Minimum per class	10
Maximum per class	10
Mean per class	10.0
Standard deviation	0.0

Table 2: Summary of the class distribution

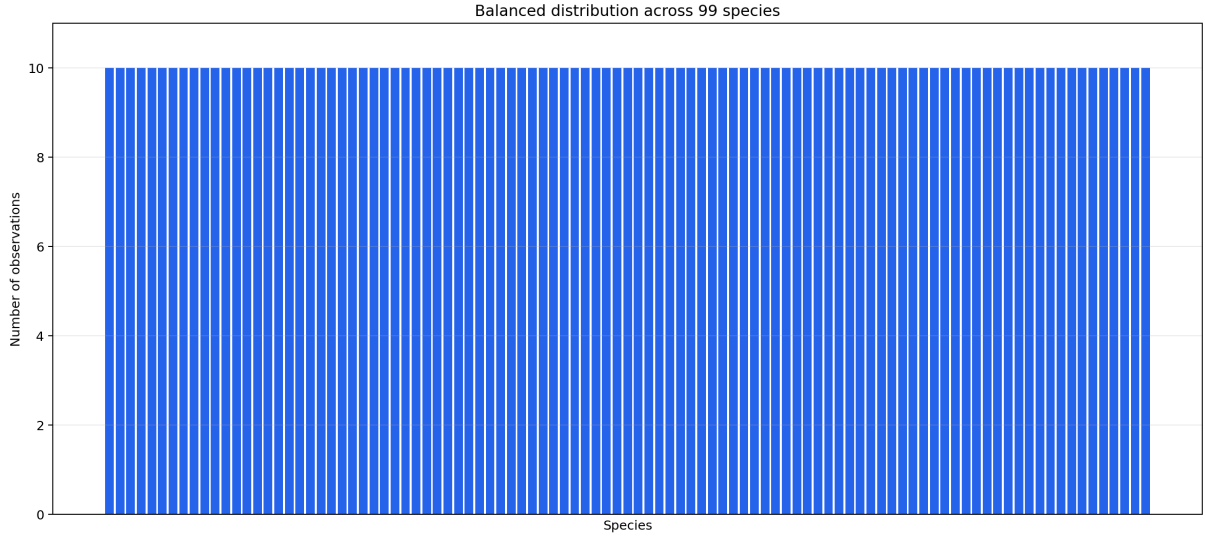


Figure 1: Number of observations per species. The uniform bar heights confirm that the training sample has no class imbalance.

2.4 Feature scale, dispersion, and distributional shape

The marginal distributions show that the three feature families operate on different scales. The **shape** variables occupy a very narrow range, with a mean standard deviation of 0.00029, whereas the **margin** and **texture** variables have mean standard deviations of 0.01688 and 0.02696, respectively. This scale heterogeneity is scientifically important because it directly affects methods that are sensitive to feature norms or to the geometry of the point cloud.

Boxplots play a central role here. Unlike a simple table of means, they simultaneously show the median, interquartile spread, skewness, and extreme values. In this dataset, they indicate that the **texture** family has the greatest dispersion, whereas **shape** is much more tightly concentrated. This observation does not establish that standardization will necessarily improve every learning procedure, but it provides a strong descriptive rationale for examining scaled variants.

Family	Min	Max	Mean Q1	Mean median	Mean Q3	Mean of means	Mean std. dev.
margin	0.00000	0.38867	0.00354	0.01028	0.02237	0.01562	0.01688
shape	0.00002	0.00301	0.00041	0.00056	0.00073	0.00061	0.00029
texture	0.00000	0.85352	0.00072	0.00543	0.01871	0.01563	0.02696

Table 3: Descriptive statistics by feature family

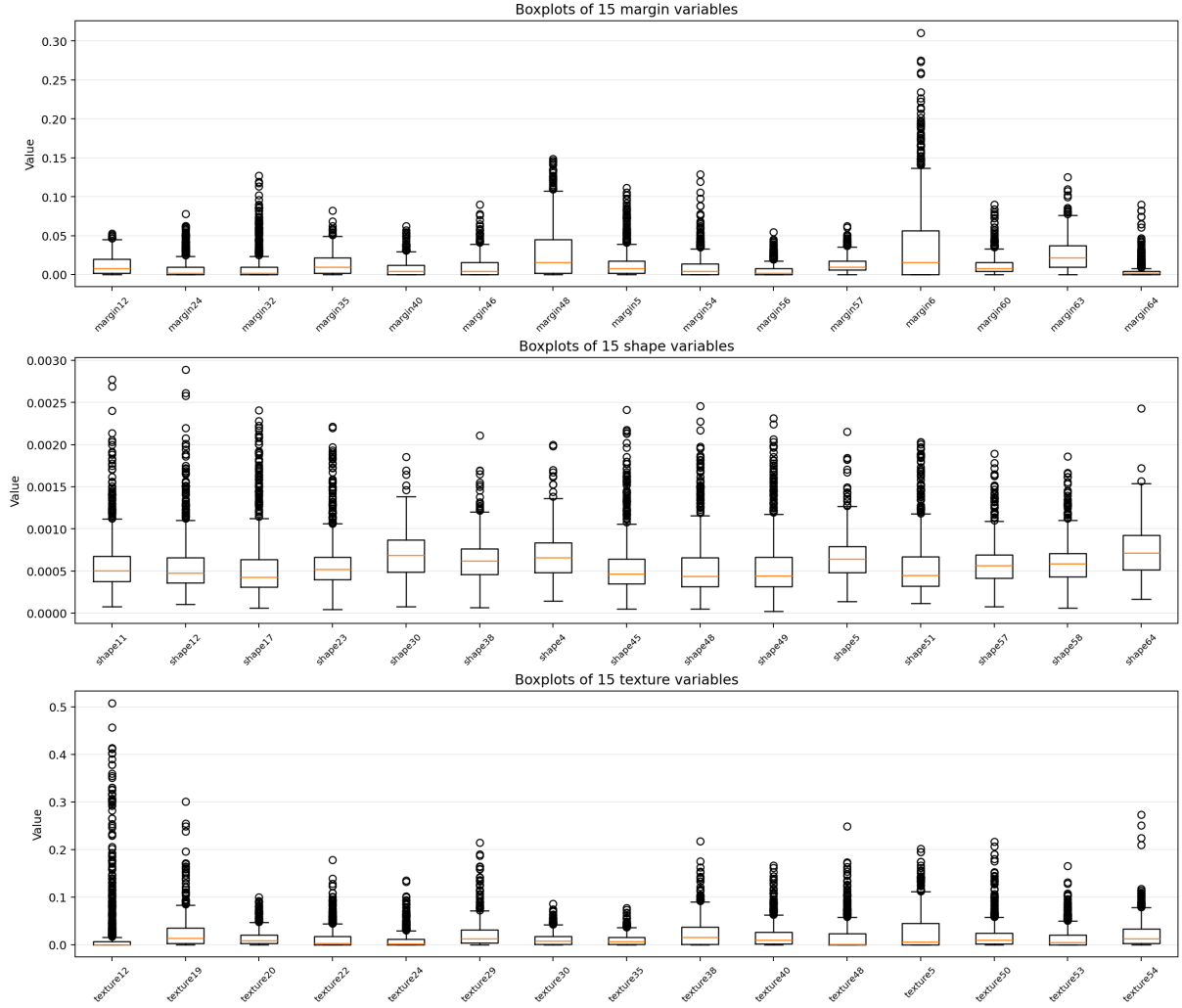


Figure 2: Boxplots of selected variables from the **margin**, **shape**, and **texture** families. The visual comparison highlights scale heterogeneity and the greater dispersion of the texture variables.

The shape of the distributions supports this interpretation. The mean absolute skewness of the 192 variables is approximately 2.20, with a maximum close to 10.91, indicating distributions that are often asymmetric and sometimes strongly concentrated near zero. This result calls for caution when interpreting mean-based statistics alone and reinforces the value of robust descriptions based on quartiles and boxplots.

2.5 Correlations and redundancy among descriptors

The correlation matrix reveals a clear block structure. The **shape** family is by far the most redundant, with a mean absolute within-family correlation of 0.6996, compared with 0.2606 for **margin** and 0.1612 for **texture**. Between-family dependencies remain weak, ranging from 0.0548 to 0.0781, suggesting that the three families do not capture exactly the same descriptive information.

Block	Mean absolute correlation
shape-shape	0.6996
margin-margin	0.2606
texture-texture	0.1612
margin-shape	0.0675
margin-texture	0.0781
shape-texture	0.0548

Table 4: Mean absolute within- and between-family correlation

Rank	Pair	Correlation
1	shape49 - shape50	0.9939
2	shape17 - shape18	0.9939
3	shape15 - shape16	0.9937
4	shape16 - shape17	0.9936
5	shape18 - shape19	0.9936

Table 5: Most highly correlated variable pairs

In total, 445 variable pairs have an absolute correlation of at least 0.90, and 197 have a correlation of at least 0.95. These magnitudes justify examining dimensionality reduction with PCA, without presuming that it will benefit the subsequent learning stages.

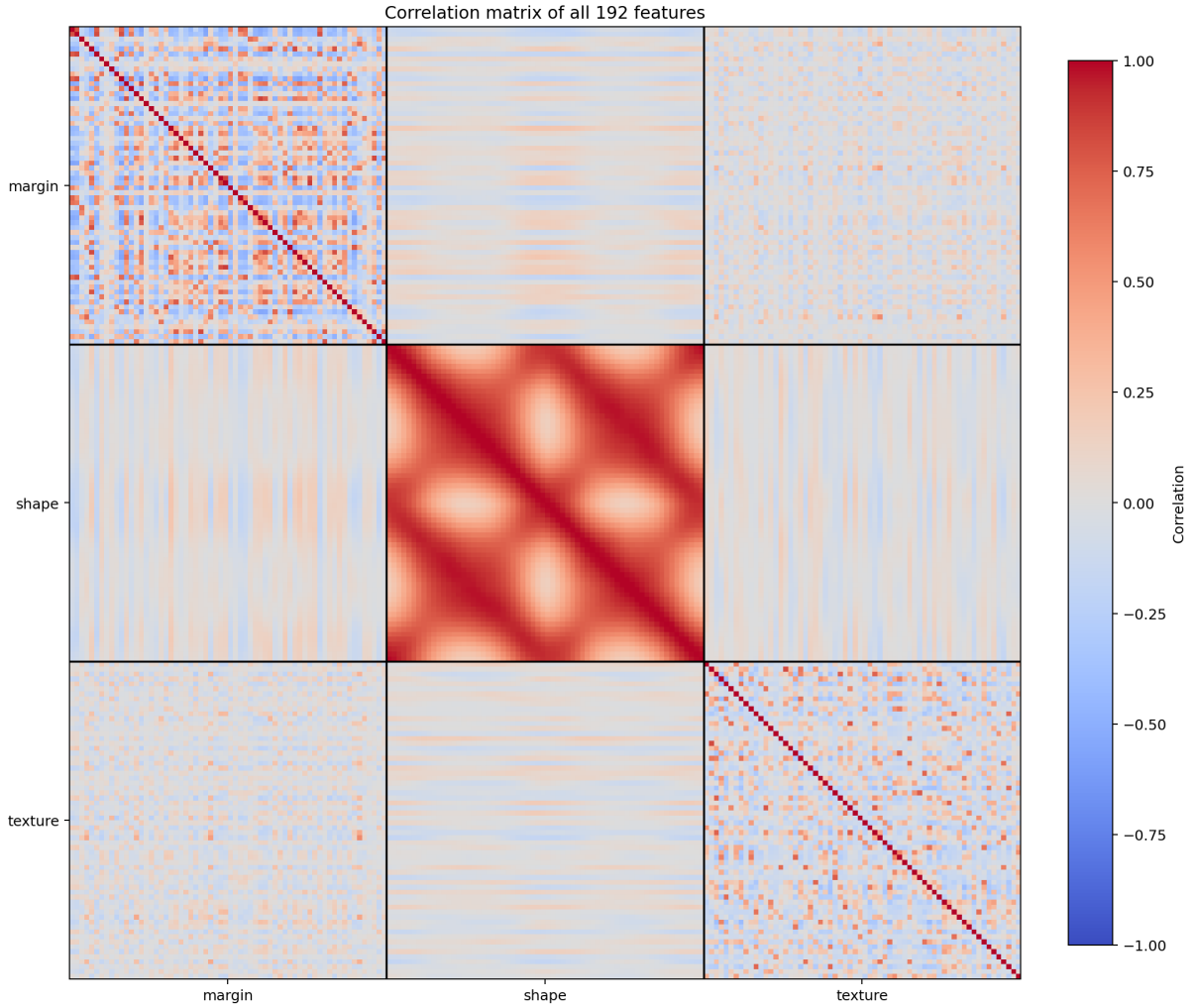


Figure 3: Correlation matrix for the 192 variables. The block structure reveals particularly strong internal redundancy within the **shape** family, while relationships between families remain weaker.

2.6 Dimensionality reduction and global organization

PCA was applied to the standardized data to summarize the dataset’s linear structure. The first two components explain approximately 24.2 % and 10.4 % of the total variance, or just under 34.6 % combined. The first five components account for 50.7 %, and the first ten for 64.7 %. The reduction is therefore meaningful, but not radical enough to suggest that a very low-dimensional space would adequately summarize the information.

The cumulative variance thresholds reinforce this interpretation. It takes 23 components to reach 80 % explained variance, 44 to reach 90 %, 69 to reach 95 %, and 116 to reach 99 %. Thus, PCA is defensible as a tool for moderate compression and mitigation of linear redundancy, but not as a drastic reduction of the problem.

Cumulative explained variance	Number of components
80 %	23
90 %	44
95 %	69
99 %	116

Table 6: Number of components required at different variance thresholds

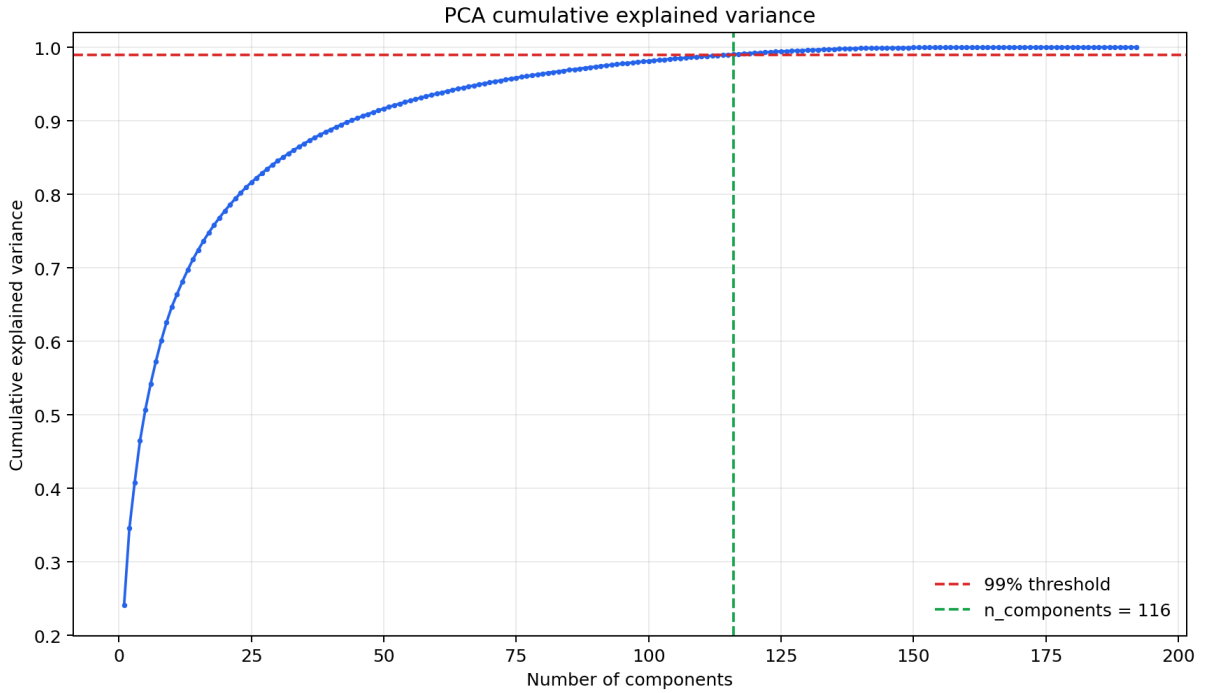


Figure 4: Cumulative explained variance after standardization. The curve’s gradual increase indicates that much of the information can be compressed, but that a substantial number of components is still required to retain nearly all variance.

The two-dimensional $\mathbf{t}$ -SNE projection complements this linear view. In the notebook, it is computed after an initial PCA reduction to 50 components, with a perplexity of 30. Its purpose is not to summarize variance quantitatively, but to examine the local consistency of neighborhoods among observations. The figure reveals partial groupings alongside overlapping regions, suggesting exploitable structure without implying trivial separation of the 99 species in a very low-dimensional space.

This interpretation must remain cautious. $\mathbf{t}$ -SNE primarily preserves local proximity relationships and depends on its hyperparameters; it provides neither a measure of global separability nor sufficient evidence to favor a particular preprocessing method. Its value here is heuristic: it confirms organization in the point cloud beyond pure noise without establishing the exact difficulty of the task.

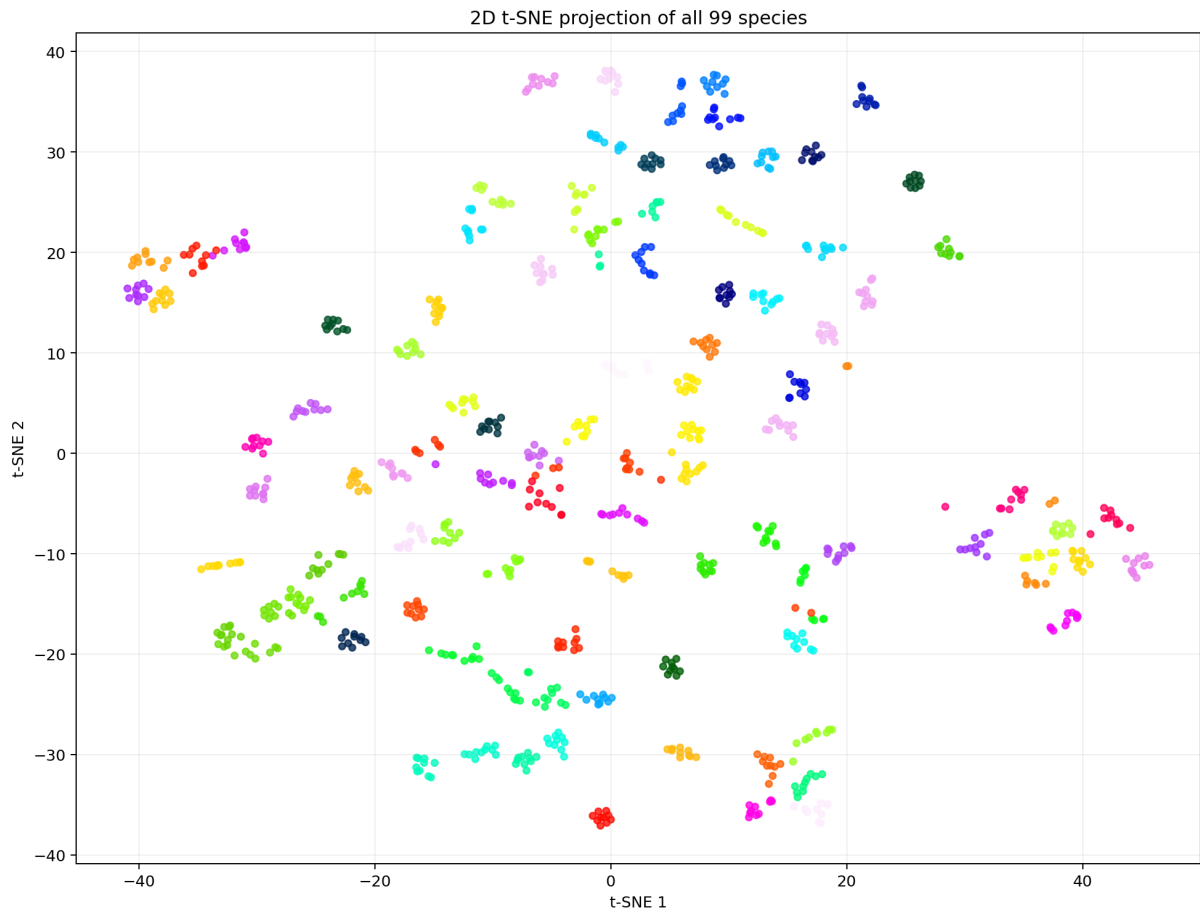


Figure 5: t -SNE 2D projection of the 99 species after standardization and an initial PCA reduction. Local clusters coexist with substantial overlap, consistent with a learnable problem that is far from trivial.

2.7 Methodological implications and summary

Four lessons emerge from this exploratory analysis. First, the dataset is structurally sound and perfectly balanced, but remains sparse at the species level, calling for restraint when interpreting variations observed on small subsets. Second, the differences in scale and dispersion among feature families provide a strong descriptive motivation for examining standardization procedures. Third, the pronounced redundancy, especially within **shape**, makes the study of PCA as a linear compression method methodologically plausible. Fourth, the reduced-dimensional projections indicate exploitable structure while emphasizing that this structure is not equivalent to simple separability.

Consequently, the preprocessing choices examined later can be scientifically motivated by this EDA, but they remain hypotheses at this stage. Standardization and PCA are reasonable exploratory options, not guarantees of improvement.

2.8 Protocol safeguards for the subsequent analysis

The preceding descriptive findings justify a cautious protocol for model comparison. The dataset is balanced, but each species remains sparsely represented; it would therefore be methodologically risky to combine variant selection, tuning, and final evaluation on

the same subset. For this reason, the remainder of the report explicitly distinguishes exploration, confirmation, and secondary corroboration.

Element	Selected value
Frozen split	80 % development / 20 % corroboration
Development-set size	792
Corroboration-set size	198
Primary cross-validation	5 outer folds
Inner tuning	3 inner folds
Primary metric	F1-macro
Secondary metric	Accuracy
Global seed	42
Status of non-tuned variants	Exploratory
Status of tuned variants	Confirmatory
Status of corroboration set	Secondary corroboration

Table 7: Safeguards in the experimental protocol

This evidence hierarchy guides the reading of all subsequent sections. The **non-tuned** variants document the effects of preprocessing and tuning, the **tuned** results determine the primary ranking, and the frozen corroboration set is used only for secondary verification.

3 Model study

This section compares six model families under a strictly common experimental framework. The primary metric is **F1-macro**, selected to give equal weight to all 99 species, while **accuracy** serves as the secondary metric.

The analysis follows three clearly distinguished levels of evidence. Exploratory variants use simple cross-validation to measure the effects of preprocessing or complexity control. The primary result for each family is the **tuned** variant, evaluated through nested cross-validation. When available, secondary corroboration on the frozen split is used solely to determine whether the signal observed on the development set persists on data never seen during modeling.

The following subsections share a common structure. Each model is first situated within the scientific rationale of the study, then examined from exploratory, confirmatory, and, where available, secondary corroboration perspectives.

The purpose of this section is not yet to reach an overall verdict among models, but to understand how each family responds to preprocessing, complexity control, and tuning. The cross-model synthesis is deliberately reserved for the next section to maintain a clear separation between family-specific analysis and the final comparative assessment.

3.1 Perceptron model

3.1.1 Role of the model

The perceptron is the study’s minimal linear baseline. Its purpose is less to compete with the best models than to quantify what an elementary decision boundary can extract from a high-dimensional multiclass problem. It therefore provides a useful reference point for measuring the effect of standardization and, more generally, the need for additional model capacity.

3.1.2 Exploration

The exploratory stage immediately shows that preprocessing plays a central role for this family. The raw variant achieves a **F1-macro** of 0.6979 and an **accuracy** of 0.7436. Adding **StandardScaler** raises **F1-macro** to 0.8228, while the **scaled_pca** variant reaches 0.8057. The main gain therefore comes from scaling; dimensionality reduction provides no additional robust improvement here.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
default	simple CV	0.6979	0.7436	0.1202	0.62
scaled_only	simple CV	0.8228	0.8460	0.1732	0.60
scaled_pca	simple CV	0.8057	0.8346	0.1923	0.65

Table 8: Comparison of perceptron variants

The perceptron therefore becomes substantially more credible after normalization, but the size of the generalization gap indicates that its room for improvement remains limited. This exploratory evidence already suggests that a low-capacity, non-probabilistic linear model will serve more as a lower bound than as a leading candidate.

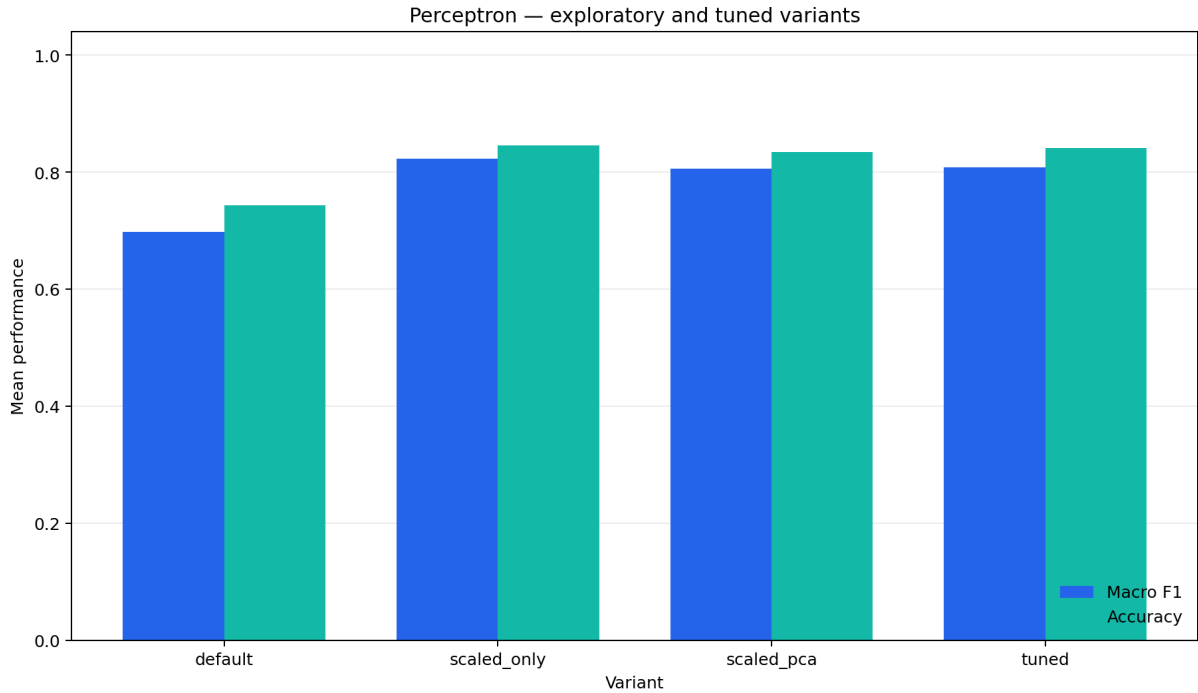


Figure 6: Exploratory comparison of perceptron variants. Nearly the entire performance increase is explained by feature standardization.

3.1.3 Confirmation

Confirmatory evaluation through nested cross-validation yields a mean **F1-macro** of 0.8085 and mean **accuracy** of 0.8409, with standard deviations of 0.0523 and 0.0411. The confirmatory score remains slightly below that of the best exploratory variant (**scaled_only**), which is consistent with a less optimistic estimate after model selection.

Confirmatory metric	Value
Mean Macro F1	0.8085
F1 standard deviation	0.0523
Mean accuracy	0.8409
Accuracy standard deviation	0.0411
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	29.79

The scientific conclusion remains stable: once the variables are standardized, the perceptron captures a substantial part of the problem’s structure, but still lags well behind the strongest families. It therefore fully serves its role as a minimal linear baseline without becoming a credible contender for the final ranking.

3.1.4 Descriptive diagnostics

The figures associated with the perceptron are primarily descriptive. The decision-function histogram shows the dispersion of margin scores, while the confusion matrix and leading-confusion summary show that residual errors remain relatively diffuse. This pattern is consistent with the limitations of a simple linear boundary on a problem with 99 classes.

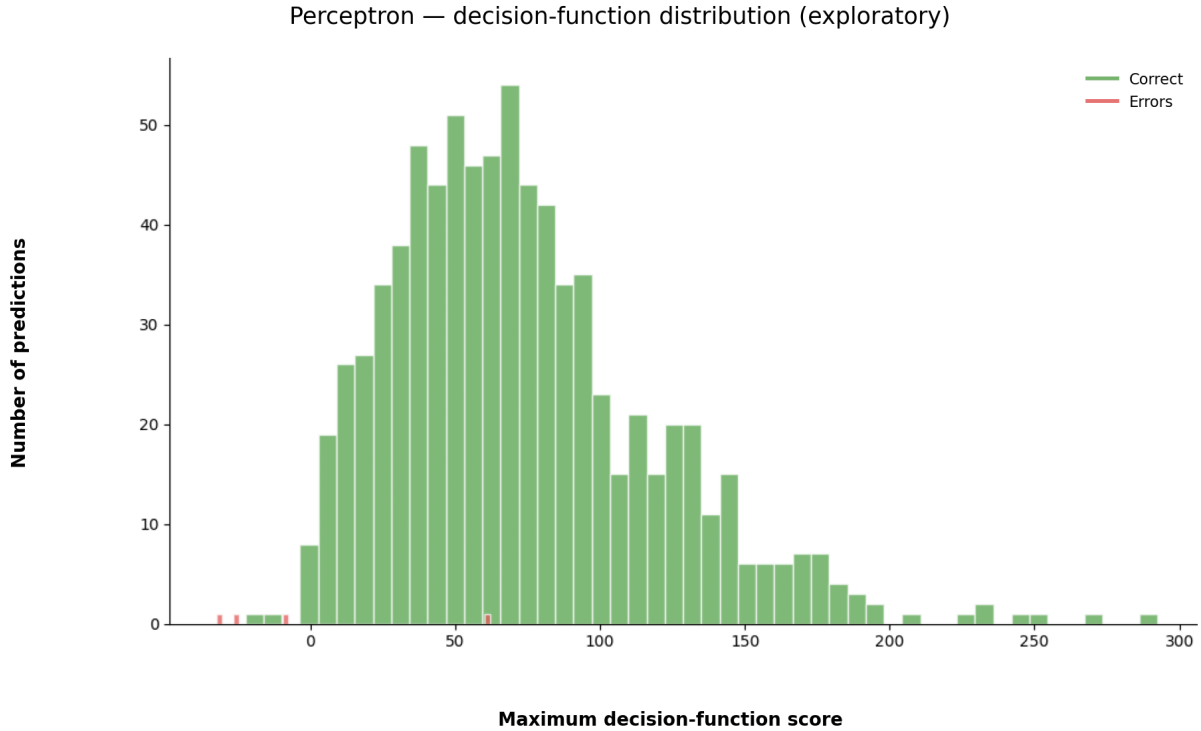


Figure 7: Exploratory histogram of the perceptron’s decision function. The score dispersion is consistent with a useful but still limited linear separation.

Perceptron — normalized OOF confusion matrix

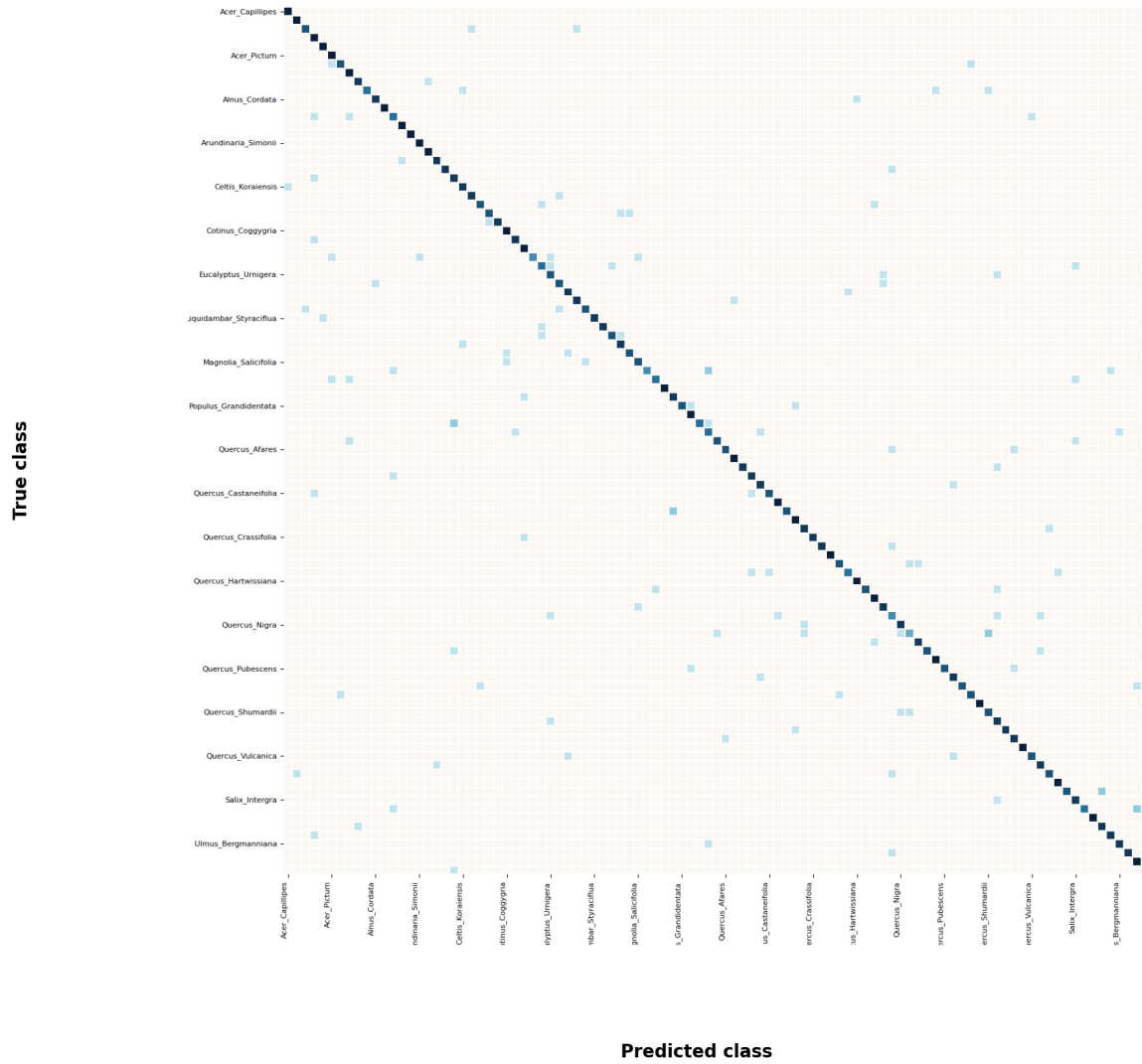


Figure 8: Normalized confusion matrix obtained from the out-of-fold predictions of the tuned variant. Residual errors remain more diffuse than for the best-performing families.

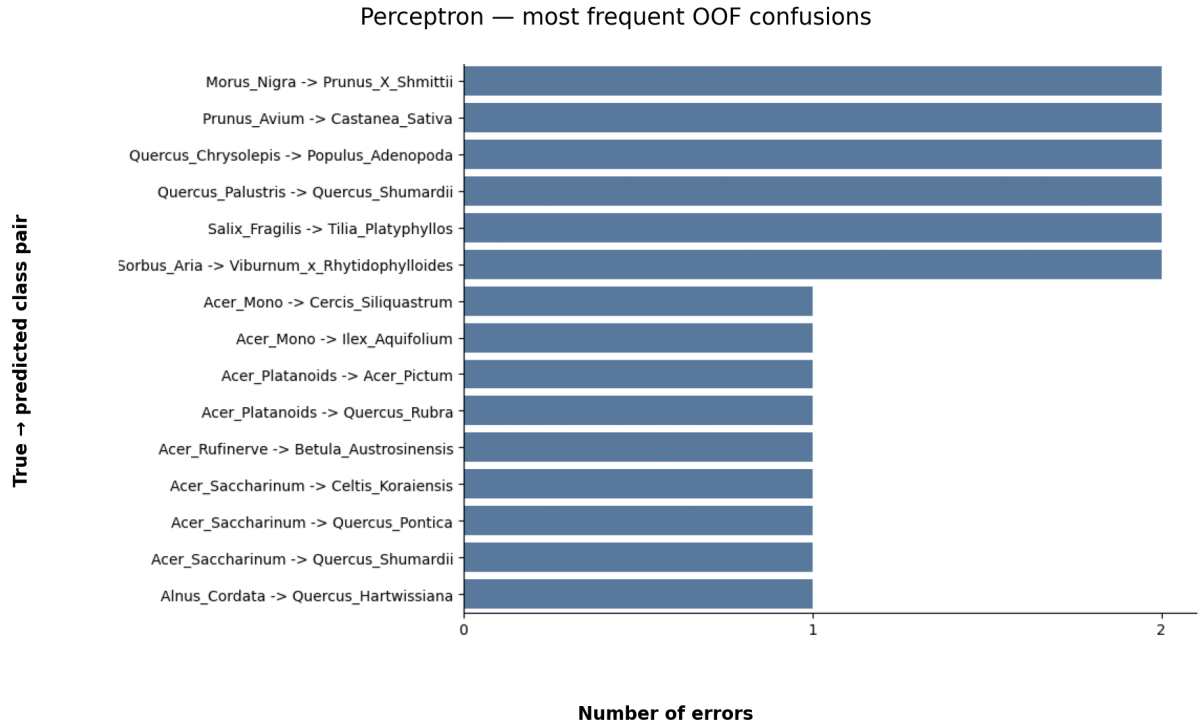


Figure 9: Leading confusions observed for the tuned perceptron. Their dispersion confirms that the model remains limited by its separation capacity.

3.1.5 Secondary corroboration

Secondary corroboration is available on the frozen split. The reference variant is `scaled_only`, selected from the exploratory results on the development set. On the corroboration set, this reference achieves a `F1-macro` of 0.8815, compared with 0.8694 for the `tuned` variant. The `tuned - reference` delta is therefore -0.0121.

Variant	Corroboration F1	Corroboration accuracy	Role
default	0.8318	0.8586	baseline
scaled_only	0.8815	0.8939	reference
scaled_pca	0.8671	0.8838	descriptive
tuned	0.8694	0.8838	tuned

For the perceptron, corroboration primarily confirms that the useful gain comes from standardization. Tuning provides no robust improvement for this inflexible family, which therefore remains chiefly an informative linear baseline.

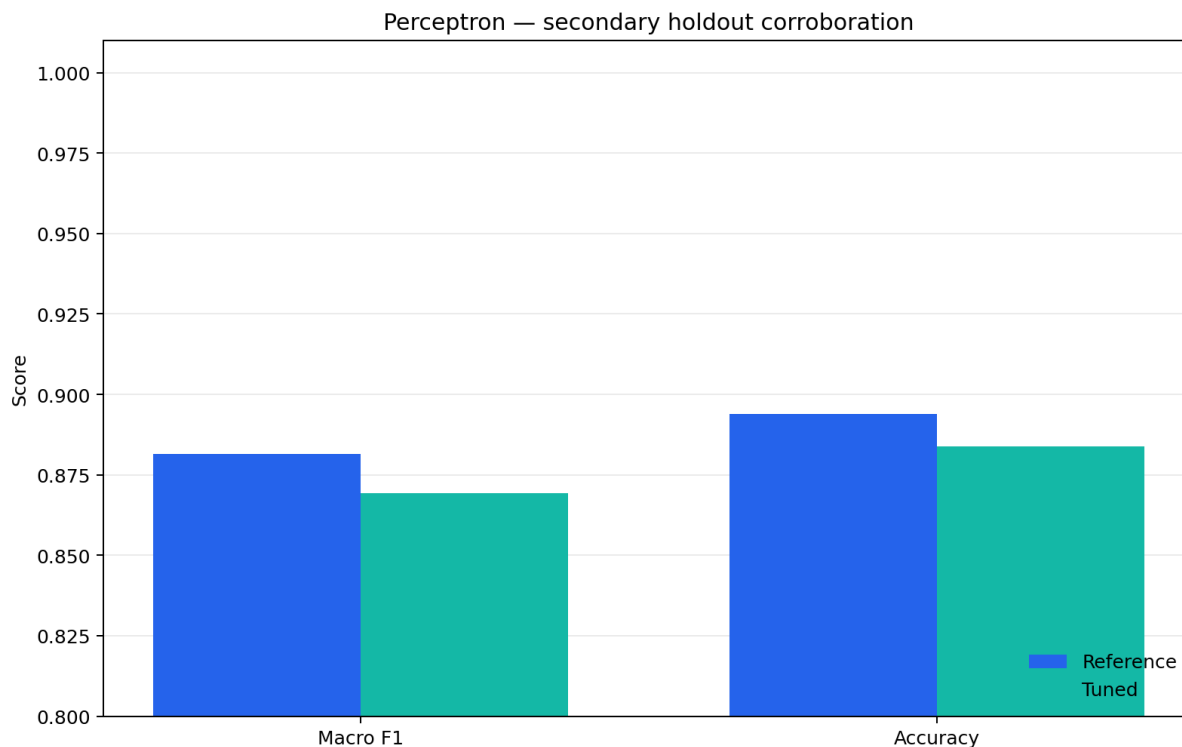


Figure 10: Secondary comparison between the reference and `tuned` perceptron variants on the corroboration set.

3.2 Logistic regression model

3.2.1 Role of the model

Logistic regression is the project’s probabilistic linear reference. It combines a simple decision structure, controlled regularization, and interpretable probabilities, making it a particularly strong anchor for assessing both predictive performance and the quality of probabilistic outputs.

3.2.2 Exploration

The exploratory results reveal a very strong dependence on scaling. Without preprocessing, logistic regression reaches only 0.2591 in F1-macro and 0.3106 in accuracy. With `StandardScaler`, it rises to 0.9798 in F1-macro and 0.9861 in accuracy. The `scaled_pca` variant remains almost level with it (0.9784 in F1-macro), with no robust gain over scaling alone.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
default	simple CV	0.2591	0.3106	0.0899	1.69
scaled_only	simple CV	0.9798	0.9861	0.0202	8.10
scaled_pca	simple CV	0.9784	0.9848	0.0216	2.21

This stage already establishes a strong result: on this dataset, a well-regularized linear model becomes extremely effective once supplied with correctly standardized variables.

The family therefore immediately emerges as one of the leading candidates in the final ranking.

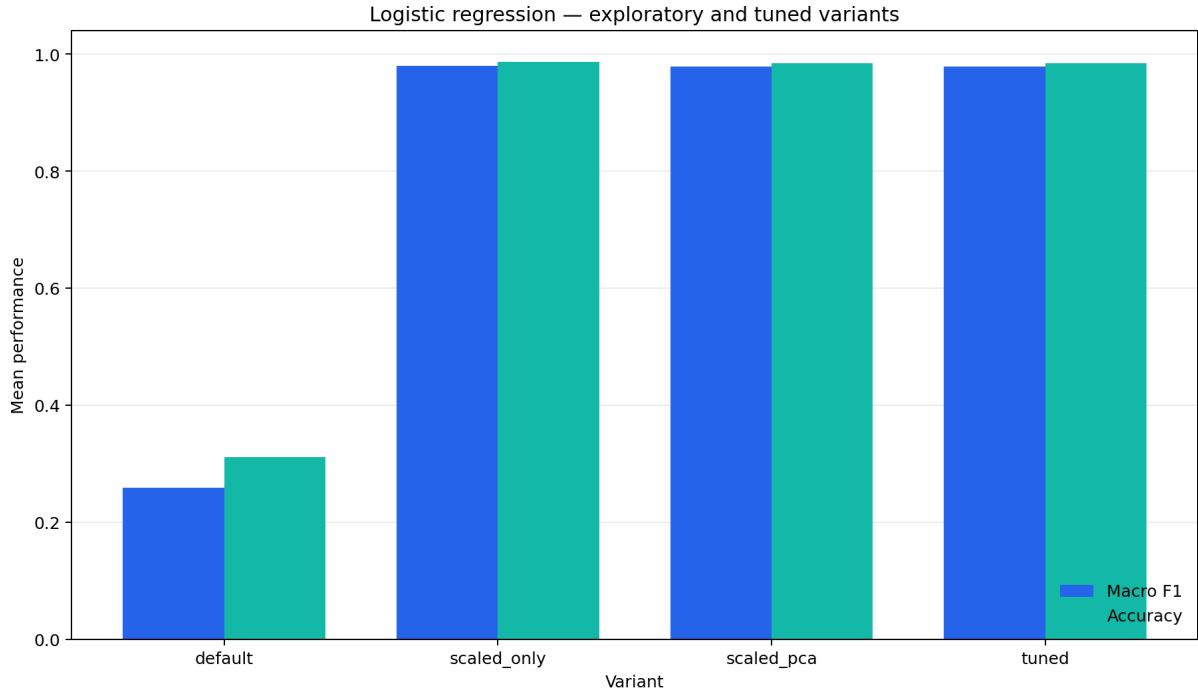


Figure 11: Exploratory comparison of logistic-regression variants. Scaling transforms a weak linear model into one of the study’s best-performing classifiers.

3.2.3 Confirmation

Nested cross-validation fully confirms this interpretation. The **tuned** variant achieves a mean F1-macro of 0.9787 and mean accuracy of 0.9848, with standard deviations of 0.0198 and 0.0146. The confirmatory estimate is virtually identical to that of the best exploratory variant, indicating excellent signal stability.

Confirmatory metric	Value
Mean Macro F1	0.9787
F1 standard deviation	0.0198
Mean accuracy	0.9848
Accuracy standard deviation	0.0146
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	77.05

Logistic regression leads the confirmatory means and, together with the SVM, forms the report’s top pair. Its strength does not come from complex tuning, but from the

remarkable fit between linear regularization, standardization, and the structure of the problem.

3.2.4 Descriptive diagnostics

The diagnostics for this family help explain why it performs so well. The regularization path shows that performance stabilizes quickly within the useful range of C , suggesting that the data already determine a strong boundary once scaling is applied. The leading residual confusions are also very few relative to the observed performance level.

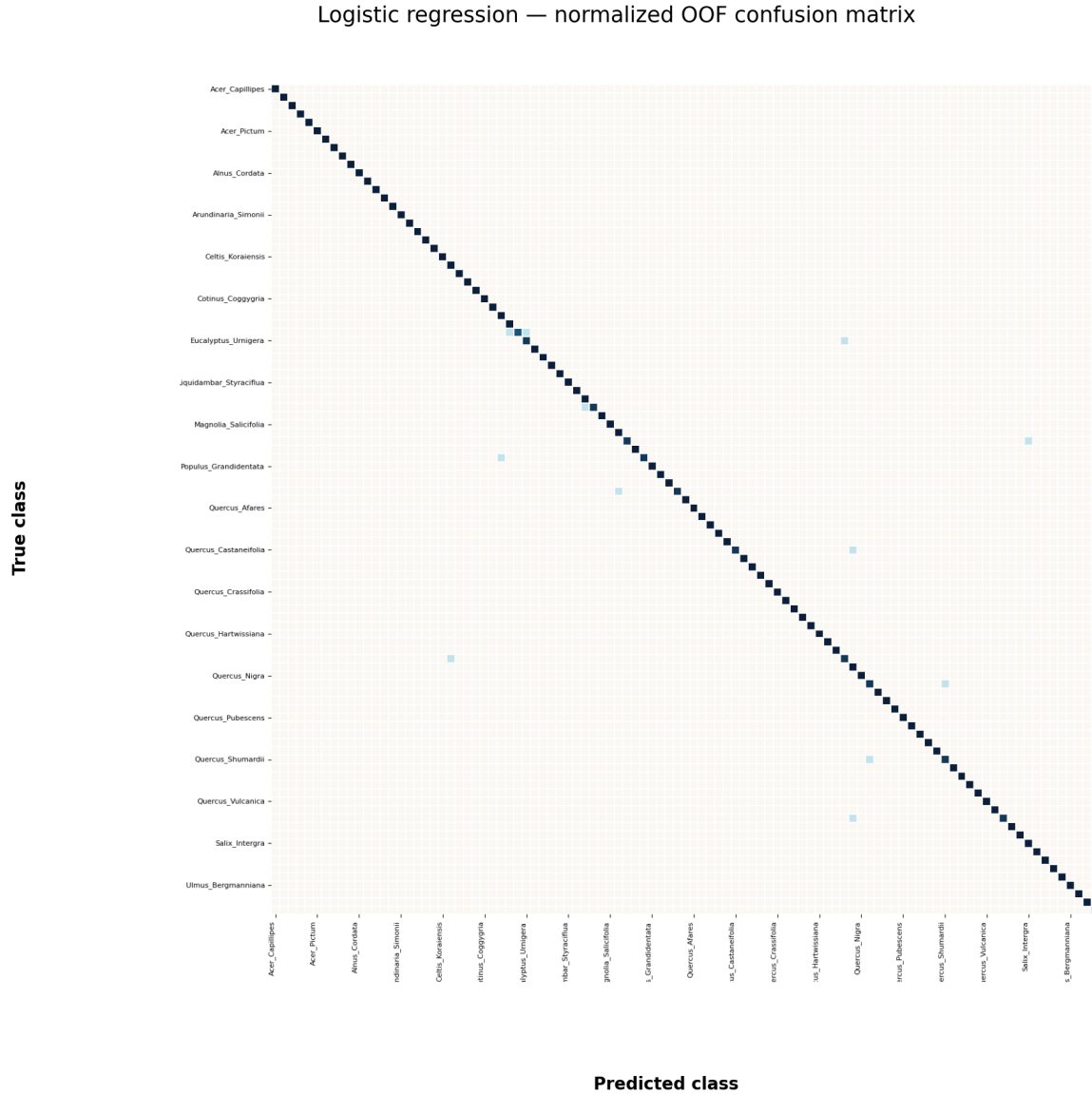


Figure 12: Normalized confusion matrix from the out-of-fold predictions of the `tuned` variant. The near-diagonal structure confirms excellent separation among classes.

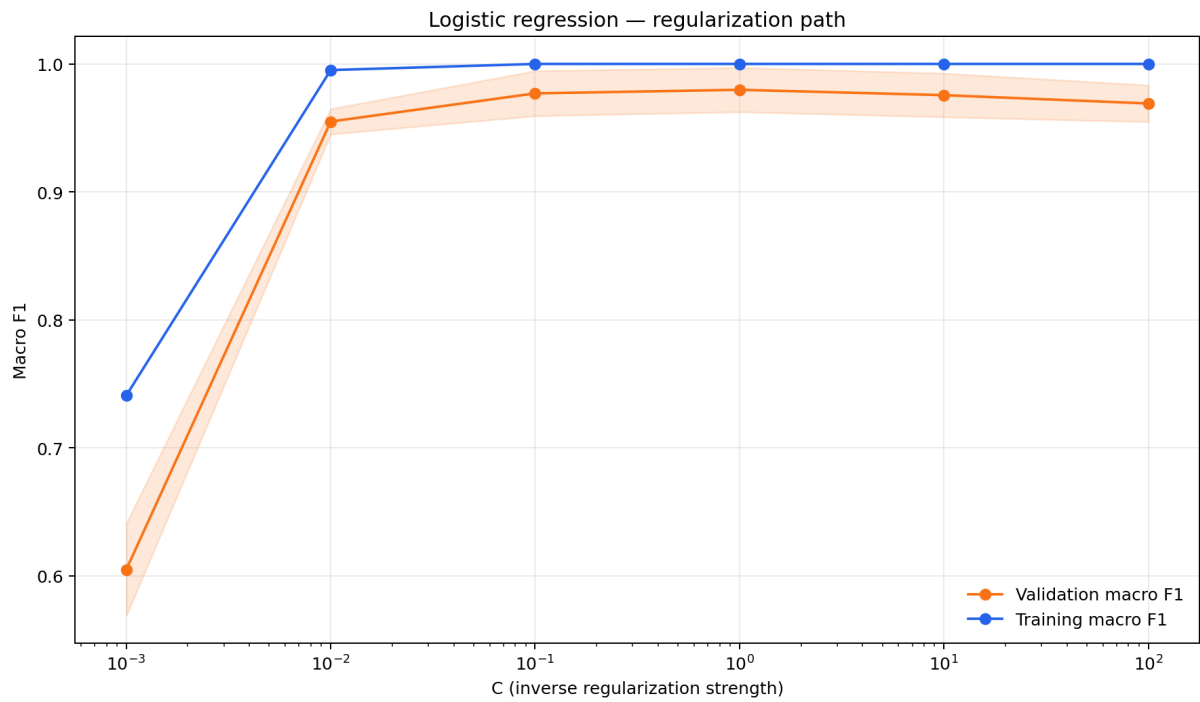


Figure 13: Exploratory diagnostic of the tradeoff between regularization and performance. This figure supports model interpretation without replacing the confirmatory estimate.

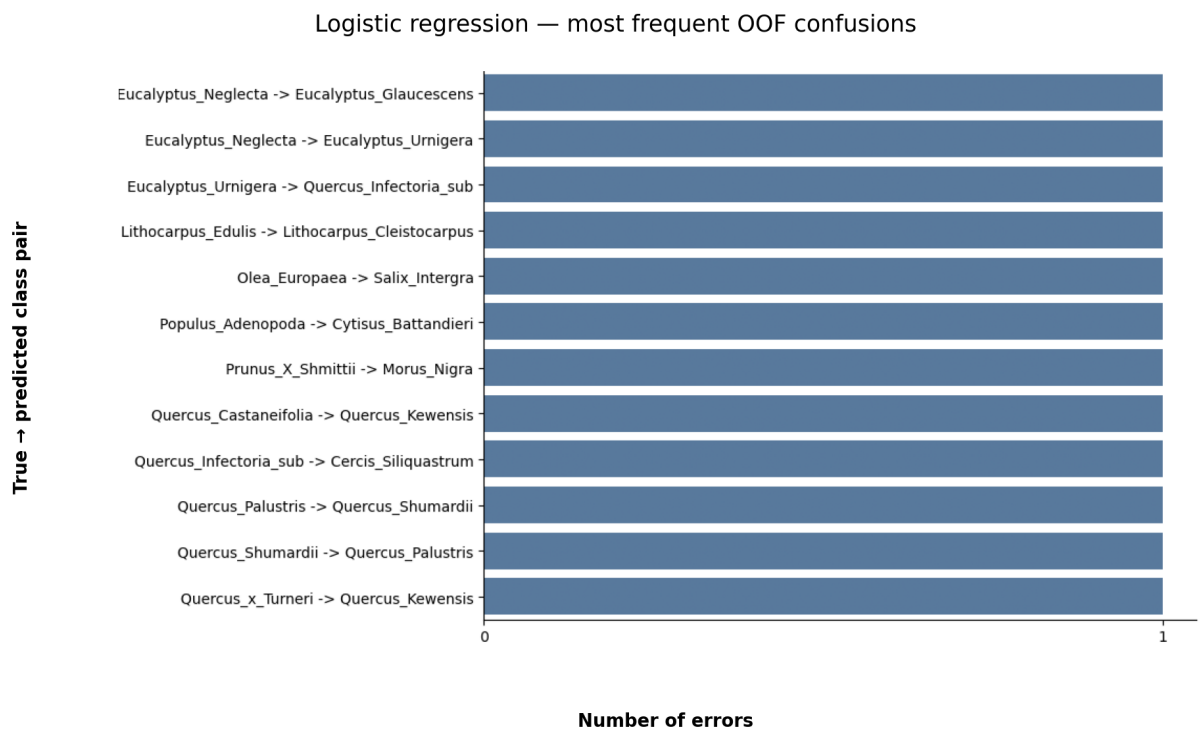


Figure 14: Leading residual confusions for logistic regression. Their small number is consistent with the observed confirmatory performance.

3.2.5 Secondary corroboration

Secondary corroboration fully supports the primary result. The `scaled_only` reference, the `scaled_pca` variant, and the `tuned` variant all achieve 0.9892 in F1-macro and 0.9899 in accuracy on the frozen split. The `tuned` - `reference` delta is therefore zero.

Variant	Corroboration F1	Corroboration accuracy	Role
default	0.4800	0.5253	baseline
scaled_only	0.9892	0.9899	reference
scaled_pca	0.9892	0.9899	descriptive
tuned	0.9892	0.9899	tuned

For logistic regression, corroboration extends a result already evident in the preceding analyses: after standardization, the family is both highly effective and remarkably stable. This combination, even more than tuning itself, makes it the report’s practical reference.

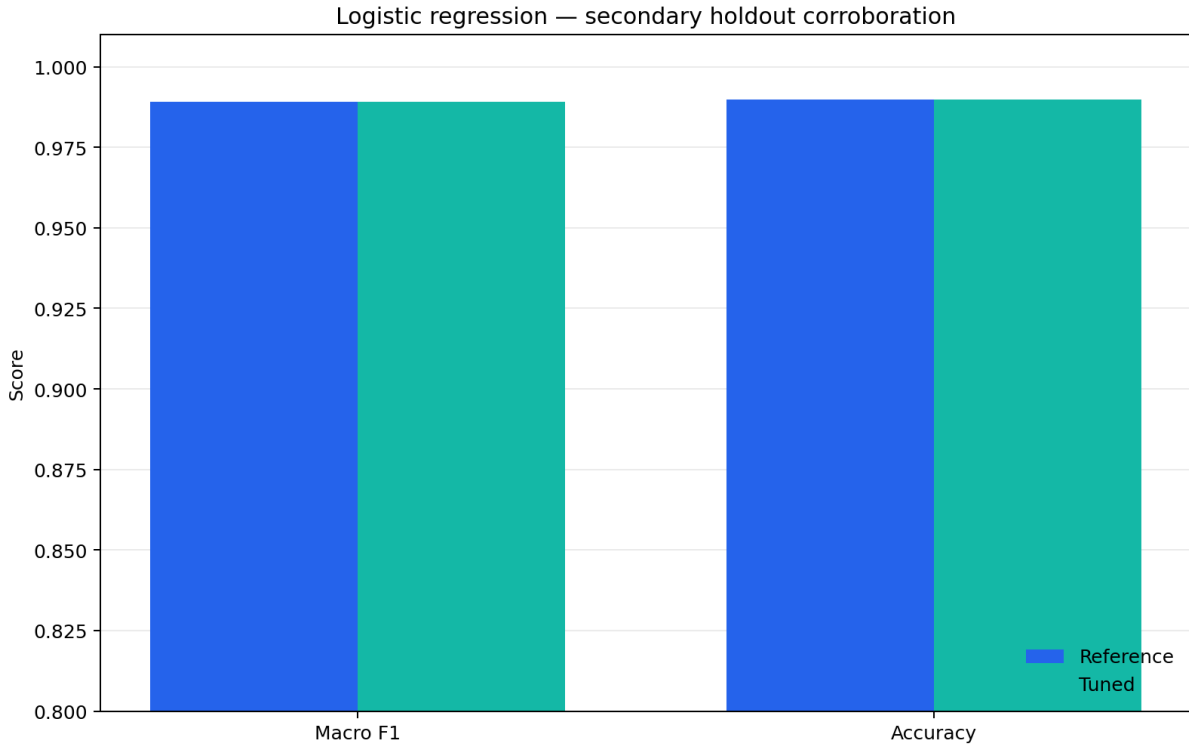


Figure 15: Secondary comparison between the reference and `tuned` logistic-regression variants on the corroboration set.

3.3 RBF SVM model

3.3.1 Role of the model

The RBF-kernel SVM is the reference nonlinear margin-based method. Its purpose is to determine whether a more flexible boundary than that of the linear models can provide a tangible gain while remaining controlled on a moderately sized dataset.

3.3.2 Exploration

The SVM is already strong without preprocessing, with a **F1-macro** of 0.8389 and an accuracy of 0.8561. Standardization nevertheless produces a clear change in scale: both **scaled_only** and **scaled_pca** reach 0.9712 in F1-macro and 0.9785 in accuracy. PCA therefore does not change the empirical interpretation, whereas scaling appears to be the decisive condition for using the RBF kernel effectively.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
default	simple CV	0.8389	0.8561	0.1364	2.36
scaled_only	simple CV	0.9712	0.9785	0.0266	2.45
scaled_pca	simple CV	0.9712	0.9785	0.0266	2.50

The SVM therefore joins the leading group during the exploratory stage. It combines very strong performance with a limited generalization gap, fully justifying its inclusion among the study’s central models.

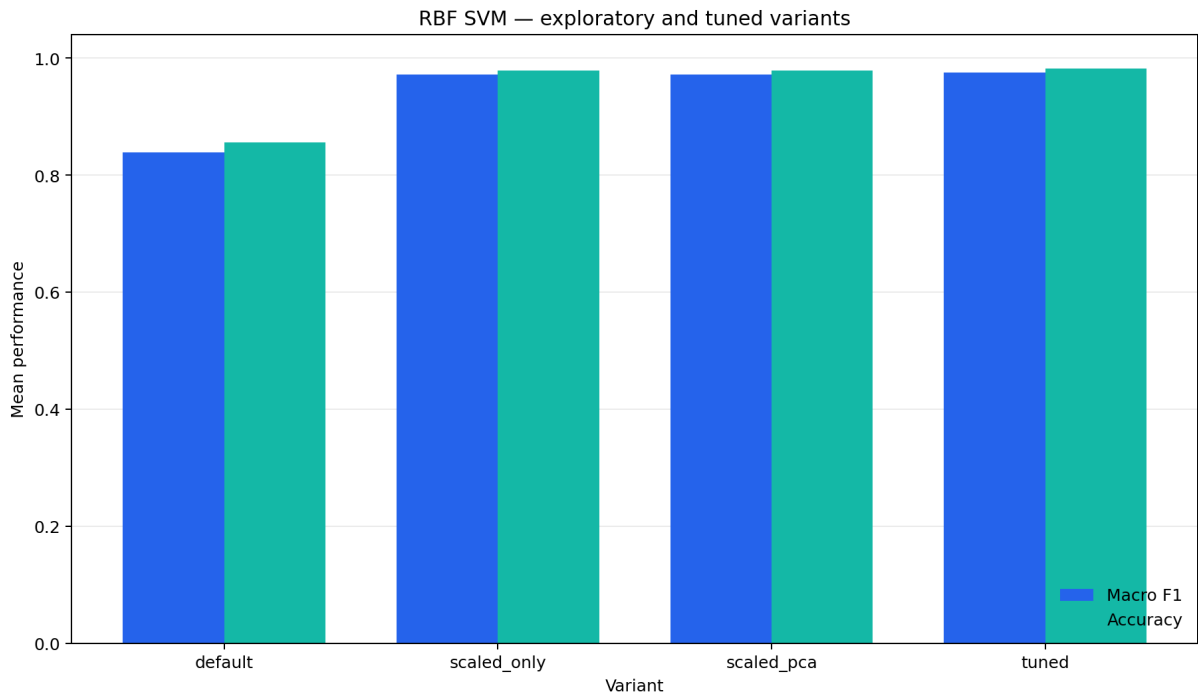


Figure 16: Exploratory comparison of SVM variants. The RBF kernel becomes fully competitive after feature standardization.

3.3.3 Confirmation

Nested cross-validation confirms this leading position. The **tuned** variant achieves a mean **F1-macro** of 0.9756 and mean **accuracy** of 0.9823, with standard deviations of 0.0207 and 0.0151. The confirmatory score is slightly higher than those of the best exploratory variants, suggesting that tuning is useful but moderate.

Confirmatory metric	Value
Mean Macro F1	0.9756
F1 standard deviation	0.0207
Mean accuracy	0.9823
Accuracy standard deviation	0.0151
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	56.49

The SVM also belongs to the leading confirmatory pair, with a mean very close to that of logistic regression. It therefore provides an excellent nonlinear alternative, at the cost of higher training time and lower structural interpretability.

3.3.4 Descriptive diagnostics

The exploratory tuning table shows that the best performance is concentrated in a compact region of the hyperparameter space. Two findings dominate. Setting `gamma = 0.1` sharply degrades the scores. In addition, PCA provides no robust gain over the standardized pipeline without dimensionality reduction. The best exploratory setting is therefore `C = 10` with `gamma = scale`, without PCA.

C / gamma	scale	0.01	0.1
1	0.9697	0.9693	0.2539
10	0.9700	0.9679	0.2885

Table 9: Exploratory SVM tuning without PCA (mean CV F1-macro)

C / gamma	scale	0.01	0.1
1	0.9697	0.9679	0.2917
10	0.9688	0.9654	0.3402

Table 10: Exploratory SVM tuning with PCA (mean CV F1-macro)

Residual confusions remain rare and localized, which is consistent with the quality of the confirmatory score.

RBF SVM — normalized OOF confusion matrix

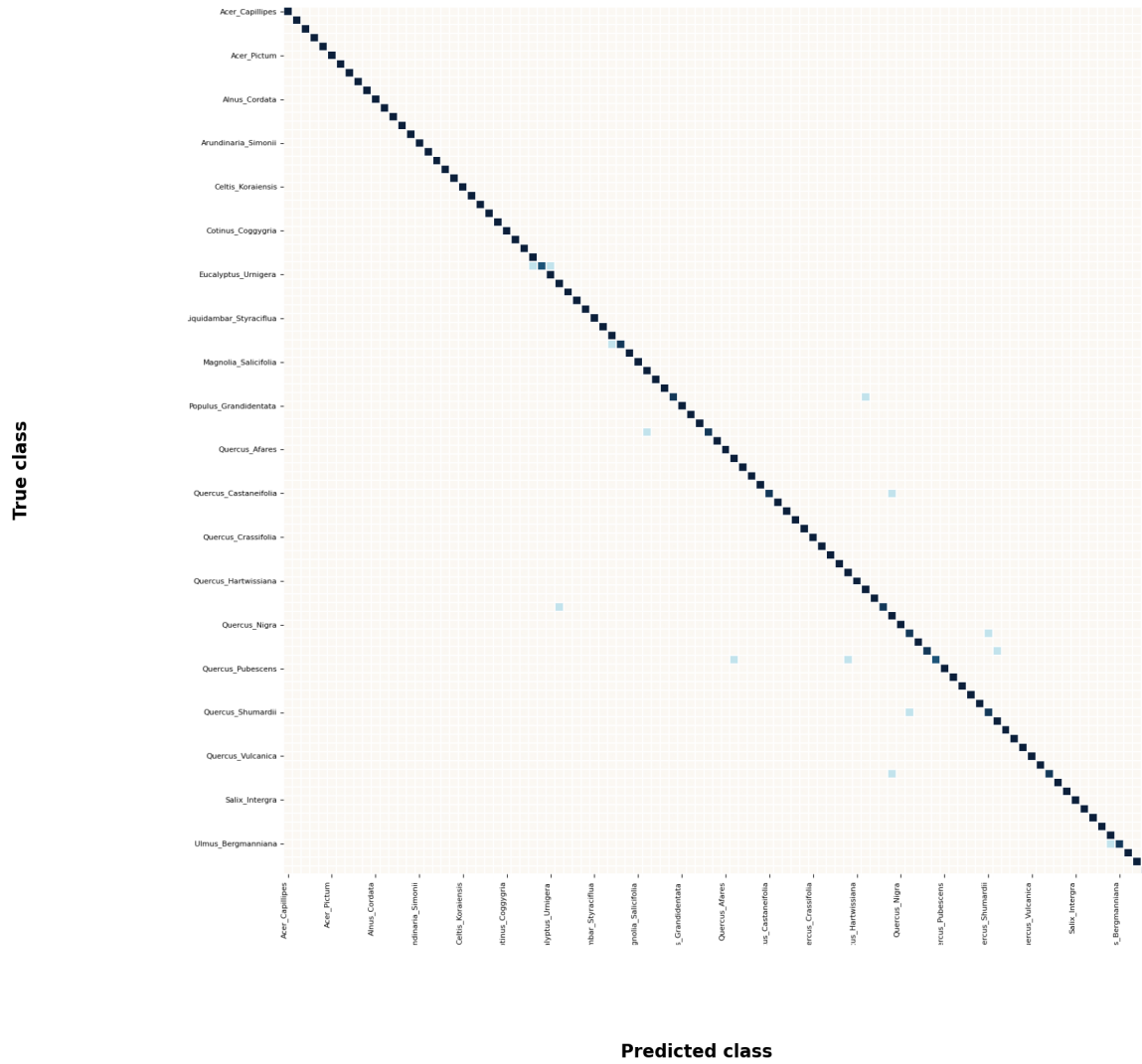


Figure 17: Normalized confusion matrix from the out-of-fold predictions of the tuned variant. Residual errors are few and concentrated.

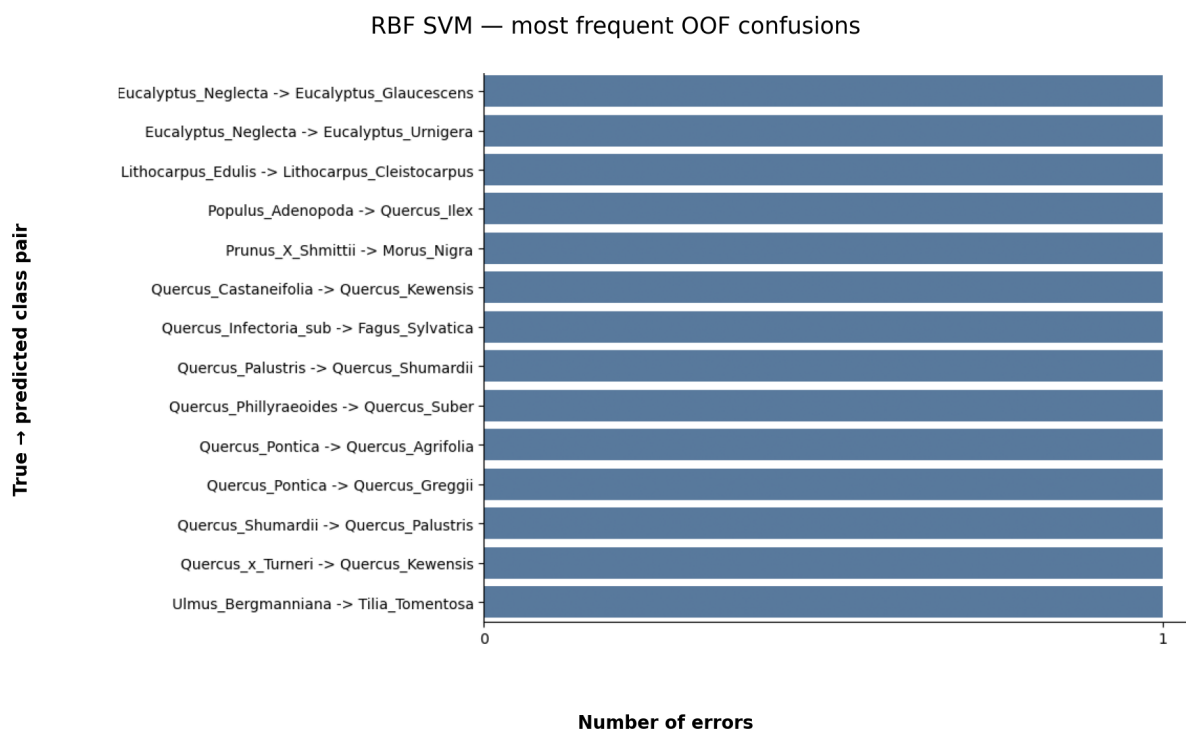


Figure 18: Leading confusions for the tuned SVM. Their limited number indicates that errors are concentrated among a few structurally difficult cases.

3.3.5 Secondary corroboration

Secondary corroboration is consistent with the confirmatory analysis and provides an even more favorable signal for tuning. The `scaled_only` reference achieves 0.9838 in F1-macro and 0.9848 in accuracy, while the `tuned` variant reaches 0.9946 and 0.9949. The `tuned` - `reference` delta is therefore +0.0108.

Variant	Corroboration F1	Corroboration accuracy	Role
default	0.9242	0.9343	baseline
scaled_only	0.9838	0.9848	reference
scaled_pca	0.9838	0.9848	descriptive
tuned	0.9946	0.9949	tuned

For the SVM, corroboration retains clear practical value: tuning remains beneficial on the frozen split. The family therefore emerges as one of the report’s most compelling solutions, very close to logistic regression in performance but more dependent on careful hyperparameter selection.

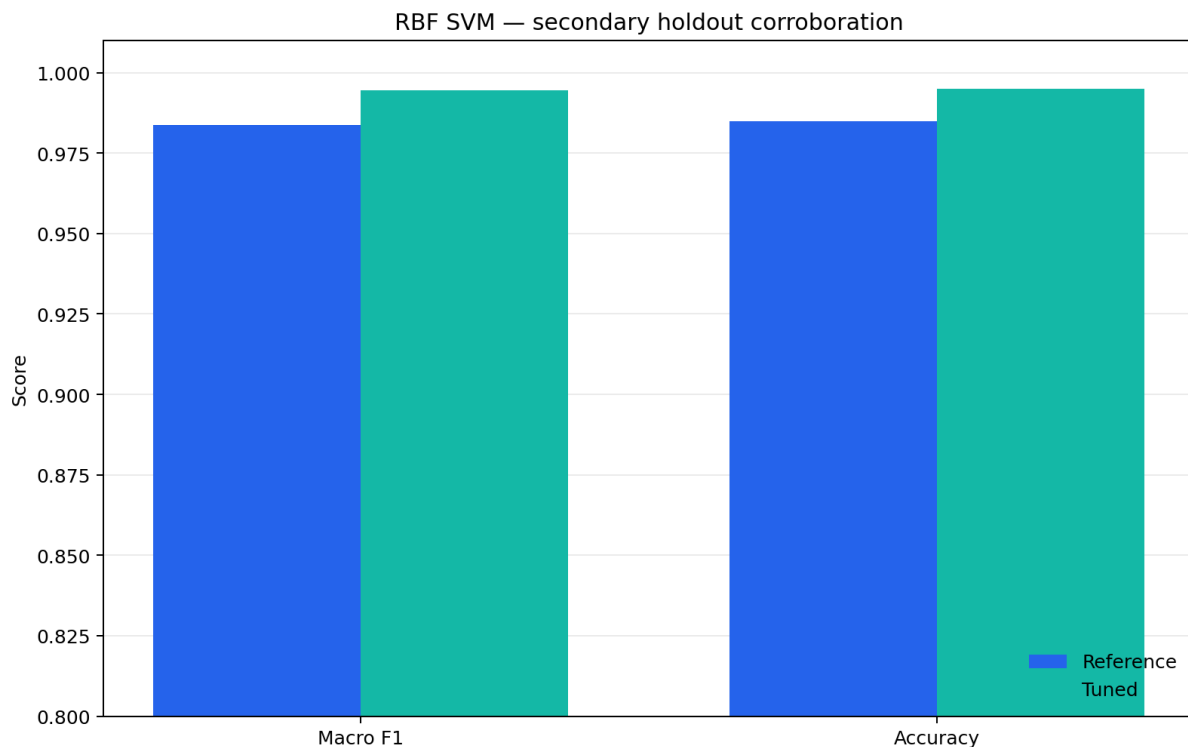


Figure 19: Secondary comparison between the reference and **tuned** SVM variants on the corroboration set.

3.4 Random forest model

3.4.1 Role of the model

The random forest represents the study’s tree-ensemble family. It is particularly relevant here because it can capture nonlinear interactions without requiring feature scaling and because it stabilizes the splits that a single decision tree struggles to generalize.

3.4.2 Exploration

The exploratory evidence is highly favorable for this family. The **default** variant achieves 0.9764 in F1-macro and 0.9785 in accuracy, with a contained generalization gap (0.0236). The **restricted_depth** variant, added as a complexity control, falls to 0.9345 in F1-macro and 0.9445 in accuracy. Restricting depth therefore slightly reduces computation time but markedly degrades performance.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
default	simple CV	0.9764	0.9785	0.0236	9.16
restricted_depth	simple CV	0.9345	0.9445	0.0651	5.42

Even at this stage, the random forest ranks among the project’s best models. Its main strength is achieving this level without sophisticated preprocessing, making it a robust and straightforward option for already informative tabular variables.

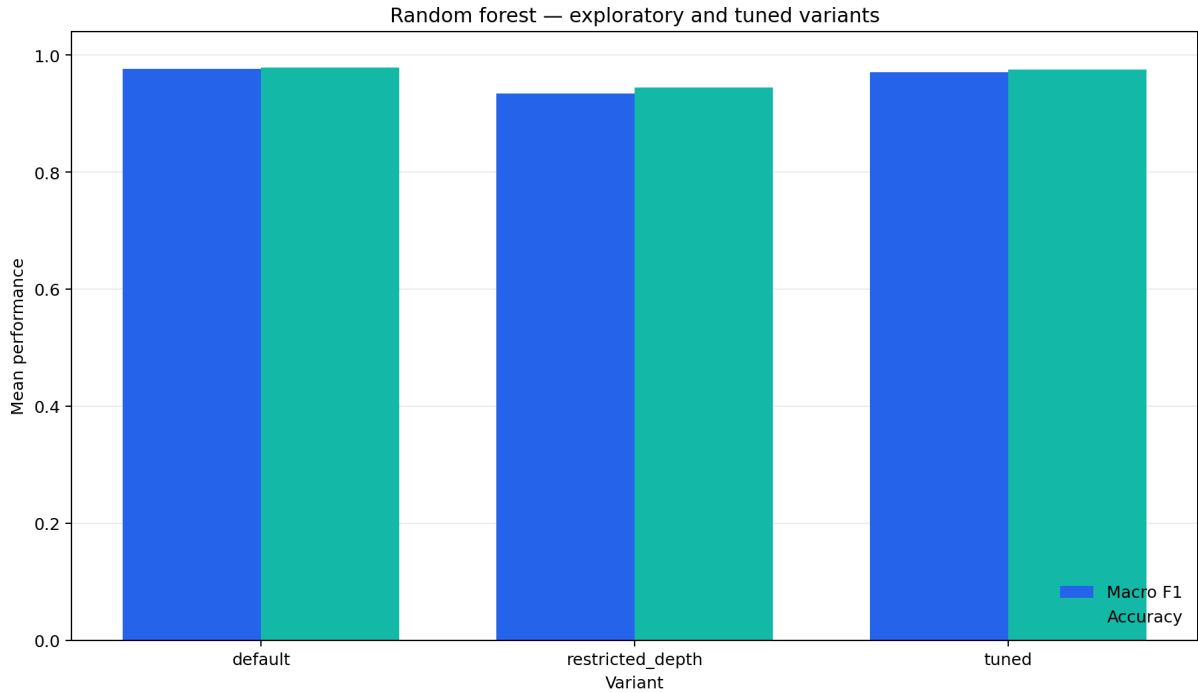


Figure 20: Exploratory comparison of random-forest variants. The default version already performs extremely well, and restricting depth proves too costly in performance.

3.4.3 Confirmation

Nested cross-validation yields a mean F1-macro of 0.9702 and mean accuracy of 0.9747, with standard deviations of 0.0140 and 0.0089. The confirmatory score remains slightly below that of the best exploratory variant, as expected when the estimate corrects for selection optimism.

Confirmatory metric	Value
Mean Macro F1	0.9702
F1 standard deviation	0.0140
Mean accuracy	0.9747
Accuracy standard deviation	0.0089
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	250.11

The random forest thus remains among the top three. It ranks slightly behind logistic regression and the SVM, but is still highly competitive and remarkably stable. Its main cost is a longer training time than the linear models.

3.4.4 Descriptive diagnostics

The OOB curve shows that marginal gains stabilize quickly as the number of trees increases. This pattern is fully consistent with the strong performance of the `default` variant and with tuning that is progressive rather than disruptive. The confusion figures further confirm that residual errors remain rare and structured.

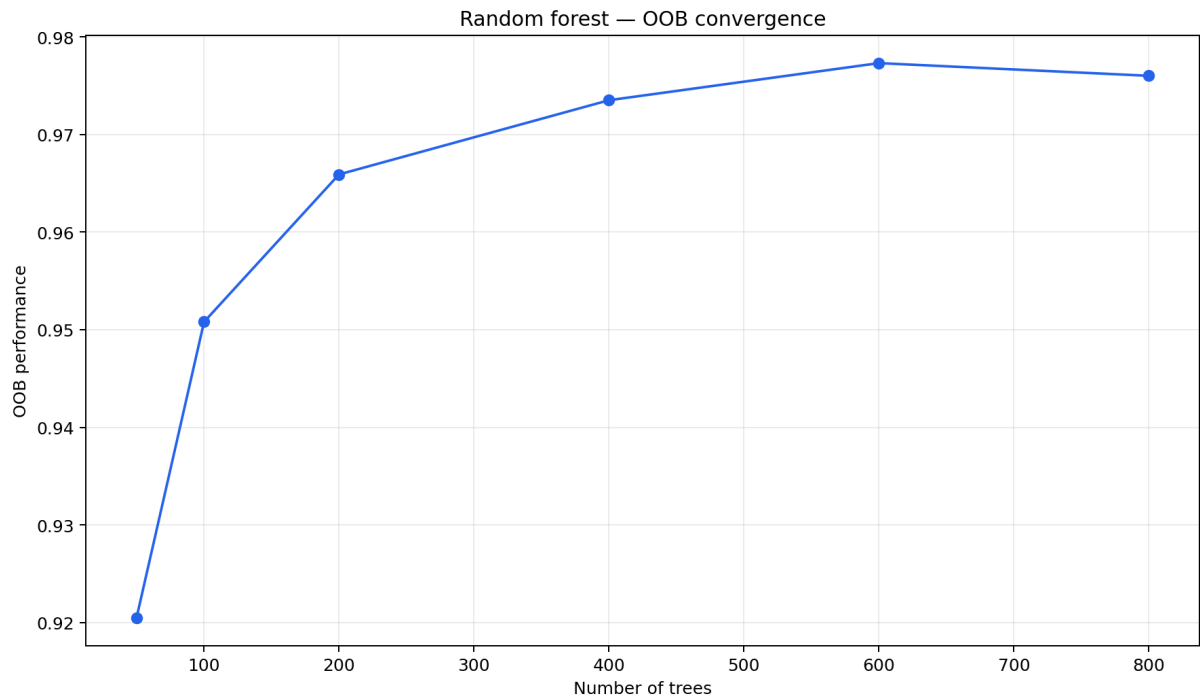


Figure 21: Exploratory evolution of the OOB score with the number of trees. The curve suggests that performance stabilizes quickly once the forest is sufficiently large.

Random forest — normalized OOF confusion matrix

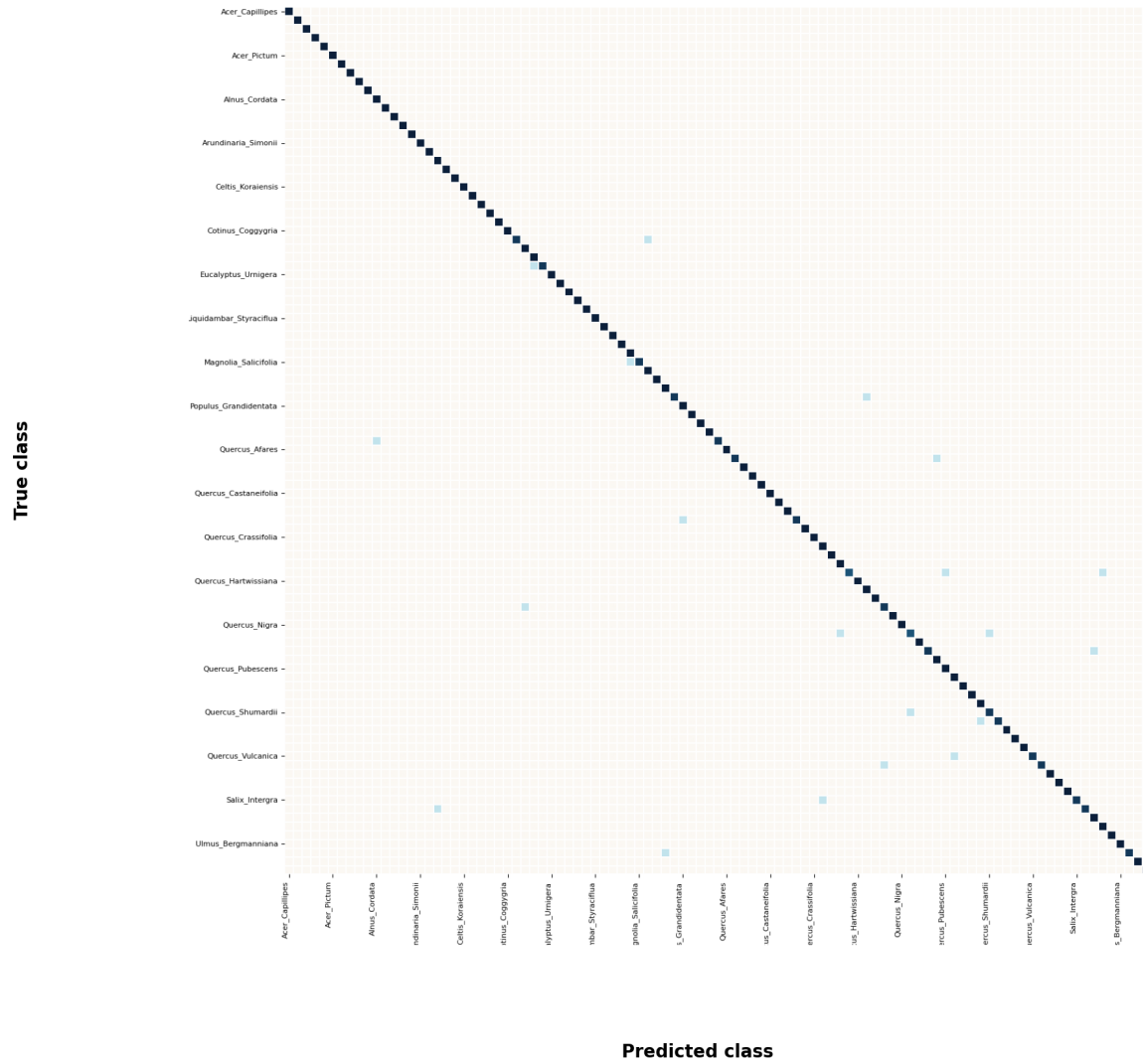


Figure 22: Normalized confusion matrix obtained from the out-of-fold predictions of the tuned variant. The pattern remains very close to that of the project's best models.

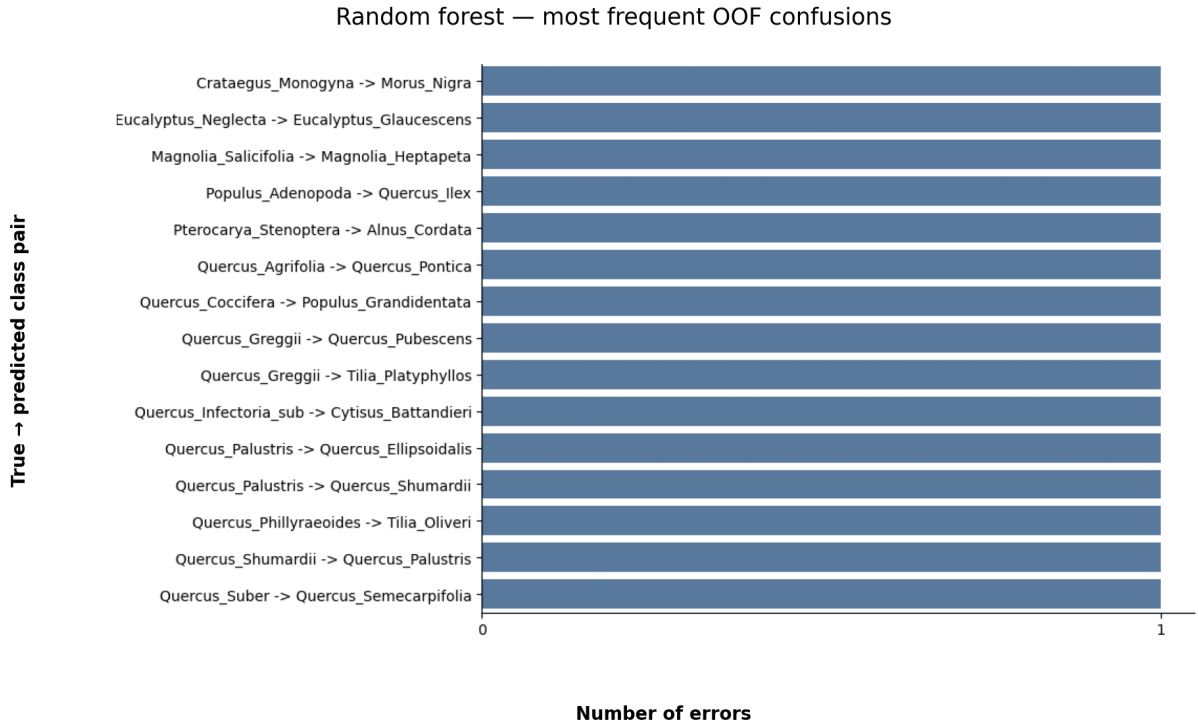


Figure 23: Leading residual confusions for the random forest. Their limited number confirms the problem’s strong effective separability for this family.

3.4.5 Secondary corroboration

Secondary corroboration is available on the frozen split. The selected reference is the `default` variant, chosen from the exploratory results on the development set. On the corroboration set, `default` and `tuned` obtain exactly the same scores: 0.9842 in `F1-macro` and 0.9848 in `accuracy`. The `tuned - reference` delta is therefore zero.

Variant	Corroboration F1	Corroboration accuracy	Role
default	0.9842	0.9848	reference
restricted_depth	0.9468	0.9545	descriptive
tuned	0.9842	0.9848	tuned

Here, corroboration confirms the family’s excellent performance without providing clear evidence of an additional gain from tuning. The random forest therefore remains a leading, robust, and highly effective model, although its default configuration is already close to the practical optimum on this dataset.

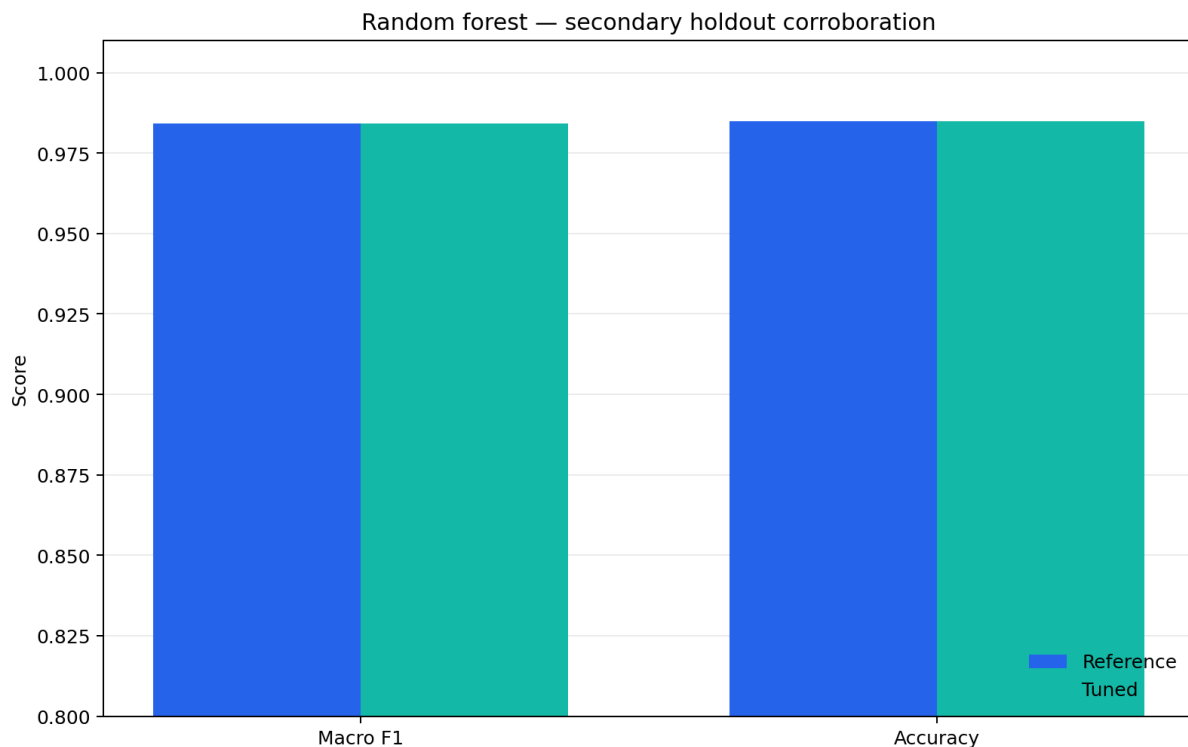


Figure 24: Secondary comparison between the reference and `tuned` random-forest variants on the corroboration set.

3.5 Decision tree model

3.5.1 Role of the model

The decision tree is studied here less as a final candidate than as a reference point for the tradeoff between interpretability and generalization. This family makes the effects of depth, leaf size, and pruning directly observable, but is also particularly prone to overfitting in a rich multiclass problem.

3.5.2 Exploration

The two exploratory variants clearly bracket the family’s behavior. The `default` variant achieves 0.5809 in F1-macro and 0.6174 in accuracy, but with a very high generalization gap (0.4191). Conversely, `restricted_depth` sharply reduces this gap (0.0797) at the cost of a collapse in performance, reaching only 0.1468 in F1-macro.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
<code>default</code>	simple CV	0.5809	0.6174	0.4191	1.04
<code>restricted_depth</code>	simple CV	0.1468	0.1529	0.0797	0.56

The exploratory conclusion is unambiguous: an overly flexible tree memorizes the data, while an overly constrained tree loses most of its discriminative power. A useful region exists for the family, but it appears narrow and unstable.

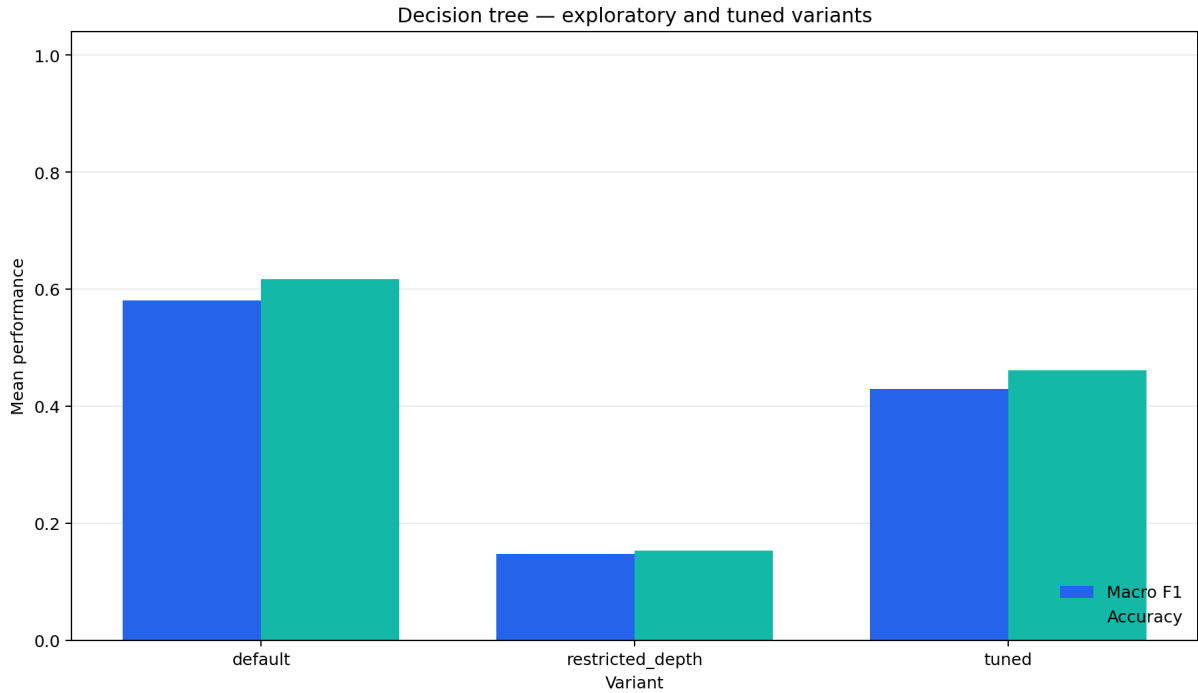


Figure 25: Exploratory comparison of decision-tree variants. The performance–complexity tradeoff remains generally unfavorable for a single tree.

3.5.3 Confirmation

Nested cross-validation confirms this fragility. The **tuned** variant achieves a mean **F1-macro** of 0.4295 and mean **accuracy** of 0.4610, with standard deviations of 0.0546 and 0.0627. The decline relative to the **default** variant does not mean that tuning degrades the model; rather, it indicates that a stricter confirmatory estimate corrects the optimism of simple evaluation for a highly unstable estimator.

Confirmatory metric	Value
Mean Macro F1	0.4295
F1 standard deviation	0.0546
Mean accuracy	0.4610
Accuracy standard deviation	0.0627
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	11.47

The decision tree therefore ranks far behind the other families. It remains pedagogically useful for examining the mechanisms of overfitting, but it is not a competitive solution in this project.

3.5.4 Descriptive diagnostics

The validation curve for `max_depth` directly exposes the core problem: the training score rises quickly with depth, whereas the validation score plateaus and then deteriorates. Accordingly, the confusion matrices show a much more diffuse pattern than those of the best-performing models.

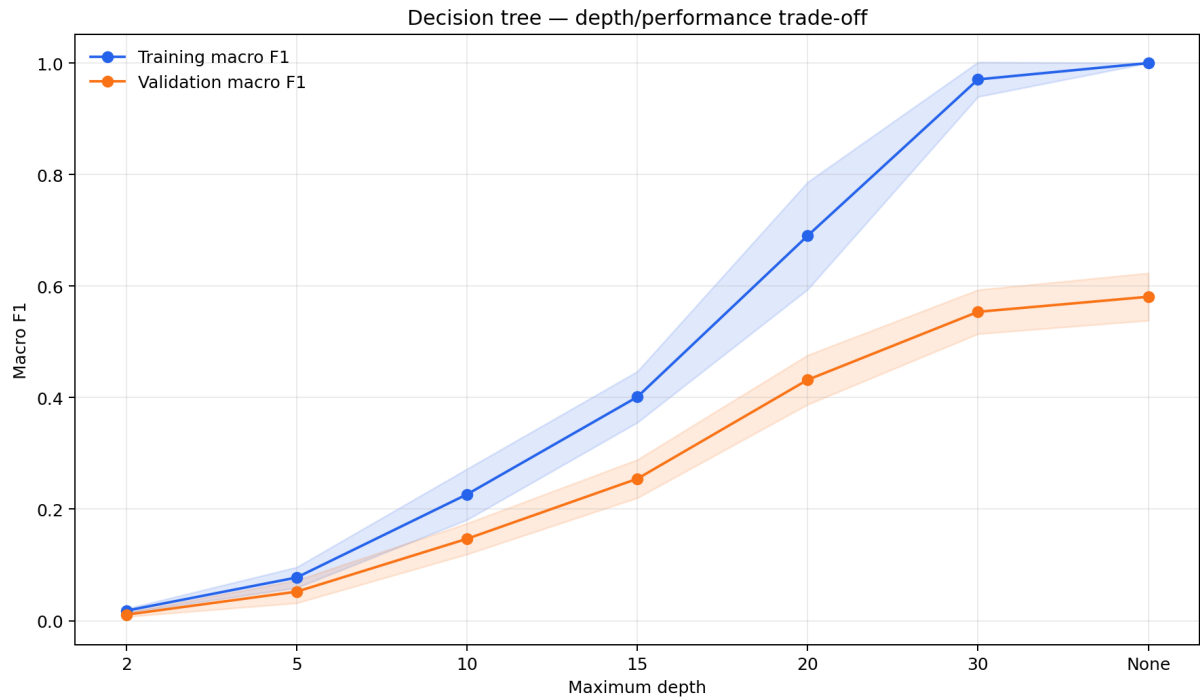


Figure 26: Exploratory validation curve for `max_depth`. The divergence between training and validation directly illustrates the family's fragility.

Decision tree — normalized OOF confusion matrix

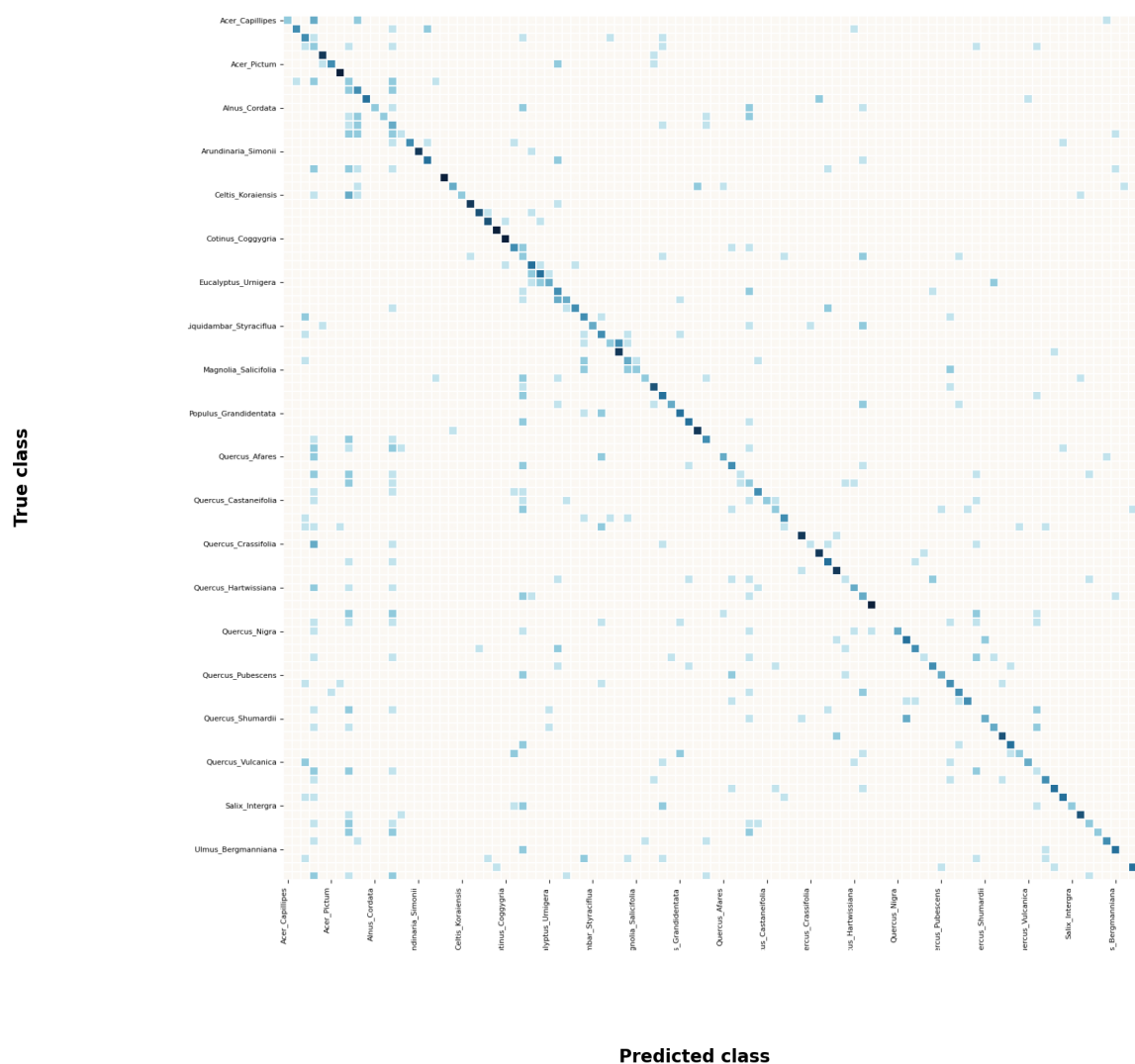


Figure 27: Normalized confusion matrix from the out-of-fold predictions of the tuned variant. Errors remain numerous and dispersed.

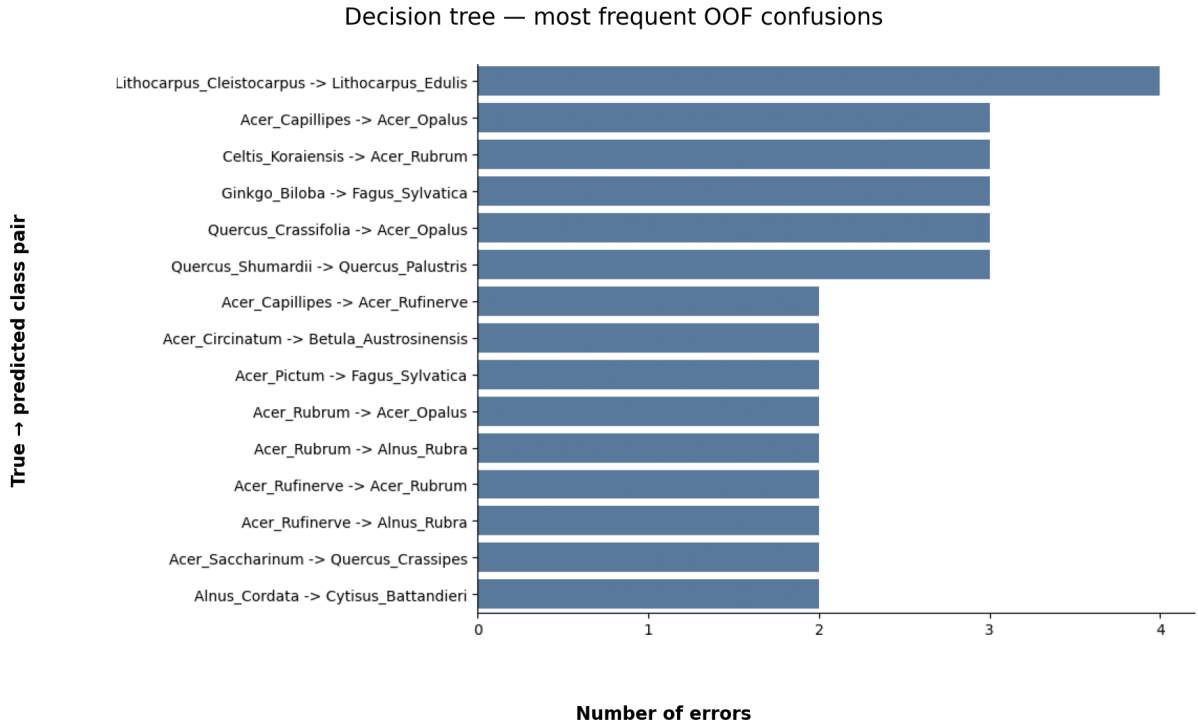


Figure 28: Leading confusions for the tuned decision tree. Their dispersion is consistent with insufficient generalization on a problem with 99 classes.

3.5.5 No secondary corroboration

Unlike the other eligible families, no secondary corroboration is published for the decision tree in the canonical artifacts. The scientific interpretation therefore rests solely on the relationship among exploration, confirmation, and confusion diagnostics.

This absence does not change the substantive conclusion. Even without secondary corroboration, the confirmatory results are sufficient to show that a single tree remains too unstable to compete with regularized linear models, the SVM, the random forest, or even the MLP.

3.6 MLP model

3.6.1 Role of the model

The MLP represents the study’s dense neural-network family. It tests whether a nonlinear capacity more flexible than linear or kernel models produces an empirical gain on a moderately sized tabular dataset with relatively few examples per class.

3.6.2 Exploration

The exploratory stage shows very clearly that preprocessing is a prerequisite for this model to function effectively. Without scaling, the MLP collapses to 0.1642 in F1-macro and 0.1863 in accuracy, with very high variability. Once the variables are standardized, the `scaled_only` variant reaches 0.9389 in F1-macro and 0.9482 in accuracy, while `scaled_pca` reaches 0.9323 and 0.9444.

Variant	Protocol	Macro F1	Accuracy	F1 gap	Time (s)
default	simple CV	0.1642	0.1863	0.0262	1.97
scaled_only	simple CV	0.9389	0.9482	0.0552	2.01
scaled_pca	simple CV	0.9323	0.9444	0.0604	1.97

The central diagnosis is therefore clear: the MLP becomes competitive only when supplied through a strict standardization pipeline. PCA provides no clear benefit and remains slightly behind the `scaled_only` variant.

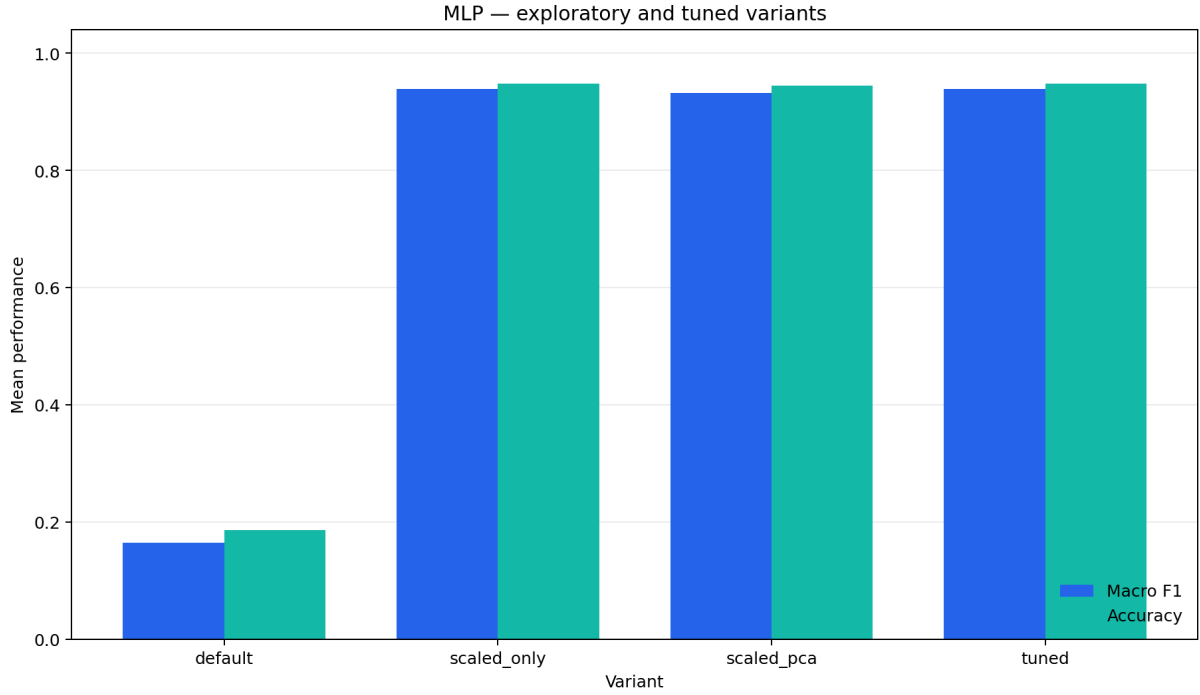


Figure 29: Exploratory comparison of MLP variants. The contrast between the raw and scaled versions illustrates the network’s dependence on feature scale.

3.6.3 Confirmation

Confirmatory evaluation yields a mean F1-macro of 0.9389 and mean accuracy of 0.9482, with standard deviations of 0.0322 and 0.0279. The `tuned` variant therefore reproduces the best exploratory result almost exactly, suggesting that tuning confirms an already strong configuration rather than providing a substantial gain.

Confirmatory metric	Value
Mean Macro F1	0.9389
F1 standard deviation	0.0322
Mean accuracy	0.9482
Accuracy standard deviation	0.0279
Protocol	nested CV
Outer folds	5
Inner folds	3
Total time (s)	38.25

The MLP thus ranks fourth in the confirmatory comparison. It remains clearly effective, but does not convert its additional flexibility into an advantage over the leading trio of logistic regression, SVM, and random forest.

3.6.4 Descriptive diagnostics

The diagnostics associated with the MLP primarily clarify its optimization dynamics. The loss curve indicates proper convergence, while the exploratory tuning table shows that most of the signal comes from hidden-layer size rather than `alpha`. The (128,) configuration consistently outperforms (64,), while PCA remains slightly behind. These diagnostics help explain the model’s behavior without competing with the confirmatory metrics.

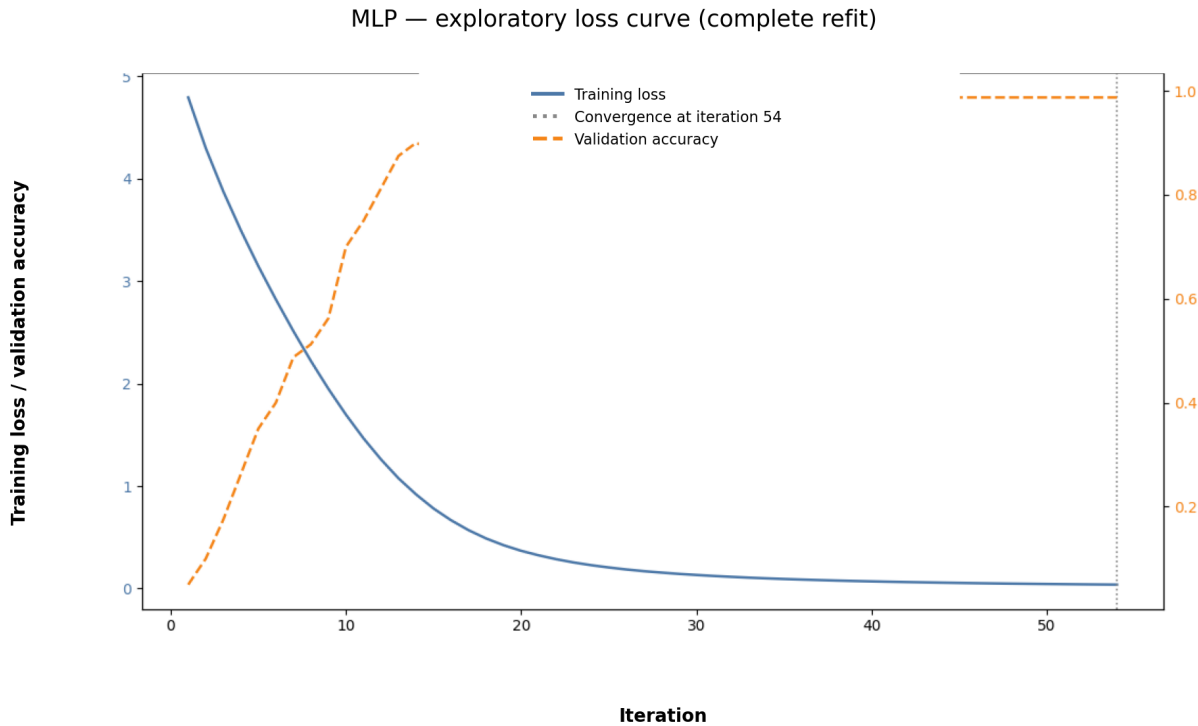


Figure 30: Exploratory diagnostic of MLP convergence. The loss stabilizes properly, although this evidence remains descriptive.

hidden_layer_sizes / alpha	0.0001	0.0010
(64,)	0.9055	0.9055
(128,)	0.9341	0.9341

Table 11: Exploratory MLP tuning without PCA (mean CV F1-macro)

hidden_layer_sizes / alpha	0.0001	0.0010
(64,)	0.9107	0.9107
(128,)	0.9266	0.9266

Table 12: Exploratory MLP tuning with PCA (mean CV F1-macro)

MLP — normalized OOF confusion matrix

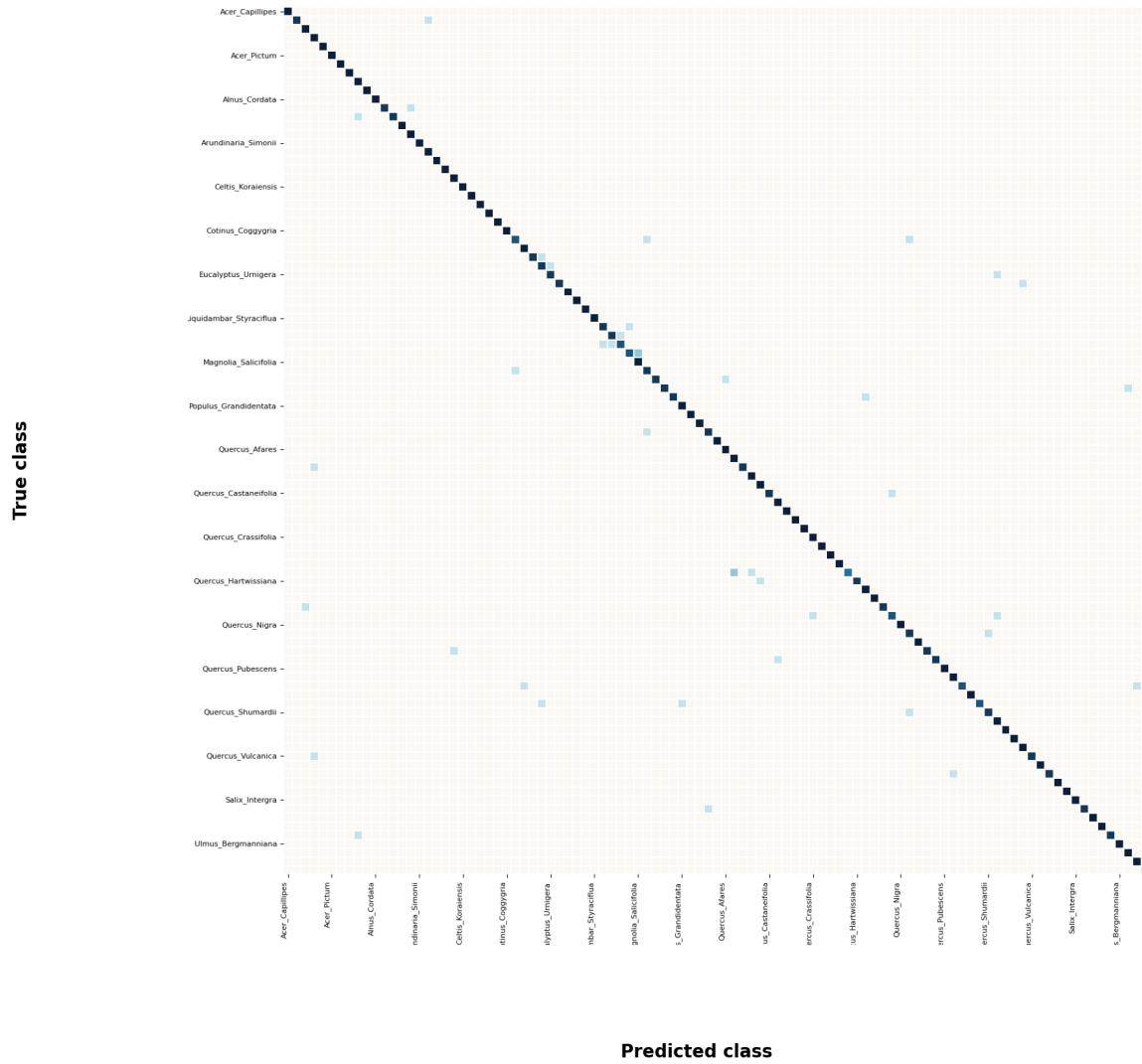


Figure 31: Normalized confusion matrix from the out-of-fold predictions of the tuned variant. Residual errors remain limited, but are more numerous than for the three best-performing families.

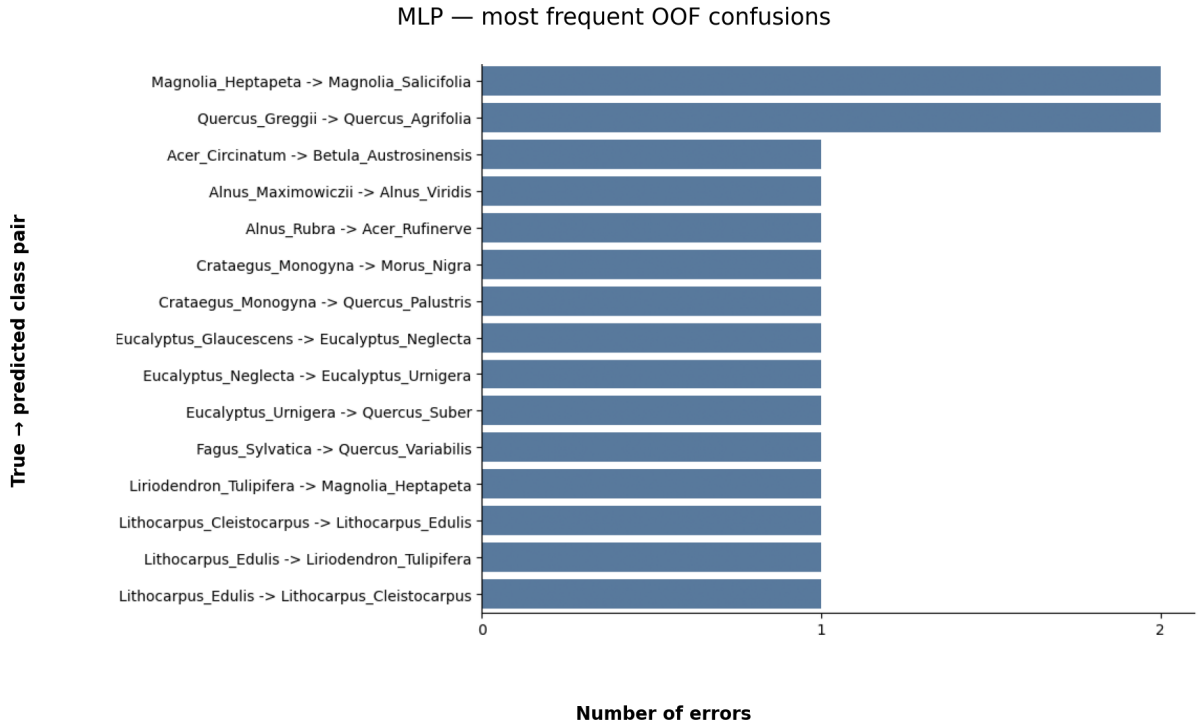


Figure 32: Leading confusions for the tuned MLP. They remain relatively concentrated and confirm an overall competitive performance level.

3.6.5 Secondary corroboration

The frozen corroboration set retains `scaled_only` as the secondary reference. On this split, the reference and `tuned` variants obtain exactly the same scores: 0.9785 in F1-macro and 0.9798 in accuracy. The `scaled_pca` variant remains slightly behind. The `tuned` - reference delta is therefore zero.

Variant	Corroboration F1	Corroboration accuracy	Role
default	0.0080	0.0404	baseline
scaled_only	0.9785	0.9798	reference
scaled_pca	0.9737	0.9747	descriptive
tuned	0.9785	0.9798	tuned

The conclusion for the MLP remains straightforward: with appropriate preprocessing, the family becomes strong, but tuning adds no robust gain on this dataset. It remains credible without surpassing the best, more parsimonious options.

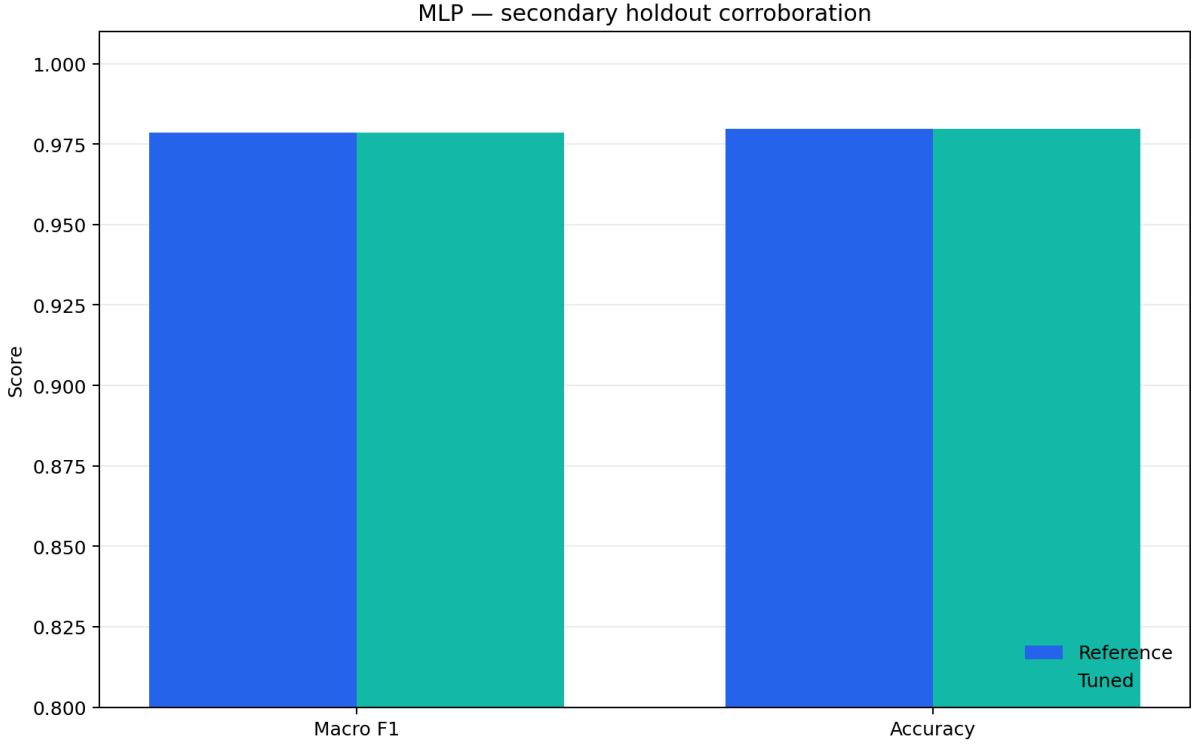


Figure 33: Secondary comparison between the reference and **tuned** MLP variants on the corroboration set.

4 Global Comparison and Discussion

The inter-model comparison presented in this section is based exclusively on the **tuned** variants evaluated through nested cross-validation. This choice limits the selection bias introduced by hyperparameter tuning, in accordance with established recommendations in the model-evaluation literature (Varma and Simon, 2006; Cawley and Talbot, 2010). The untuned variants and model-specific diagnostics retain an interpretive role, while the frozen corroboration set remains a secondary control.

4.1 Confirmatory Inter-Model Results

The following table compares the six model families on the basis of their confirmatory **tuned** performance, with **macro-F1** as the primary metric and **accuracy** as the secondary metric.

Model	Macro-F1 (mean $\pm$ std)	Accuracy (mean $\pm$ std)	Time (s)
Logistic regression	0.9787 ± 0.0198	0.9848 ± 0.0146	77.05
RBF SVM	0.9756 ± 0.0207	0.9823 ± 0.0151	56.49
Random forest	0.9702 ± 0.0140	0.9747 ± 0.0089	250.11
MLP	0.9389 ± 0.0322	0.9482 ± 0.0279	38.25
Perceptron	0.8085 ± 0.0523	0.8409 ± 0.0411	29.79
Decision tree	0.4295 ± 0.0546	0.4610 ± 0.0627	11.47

Table 13: Confirmatory inter-model comparison

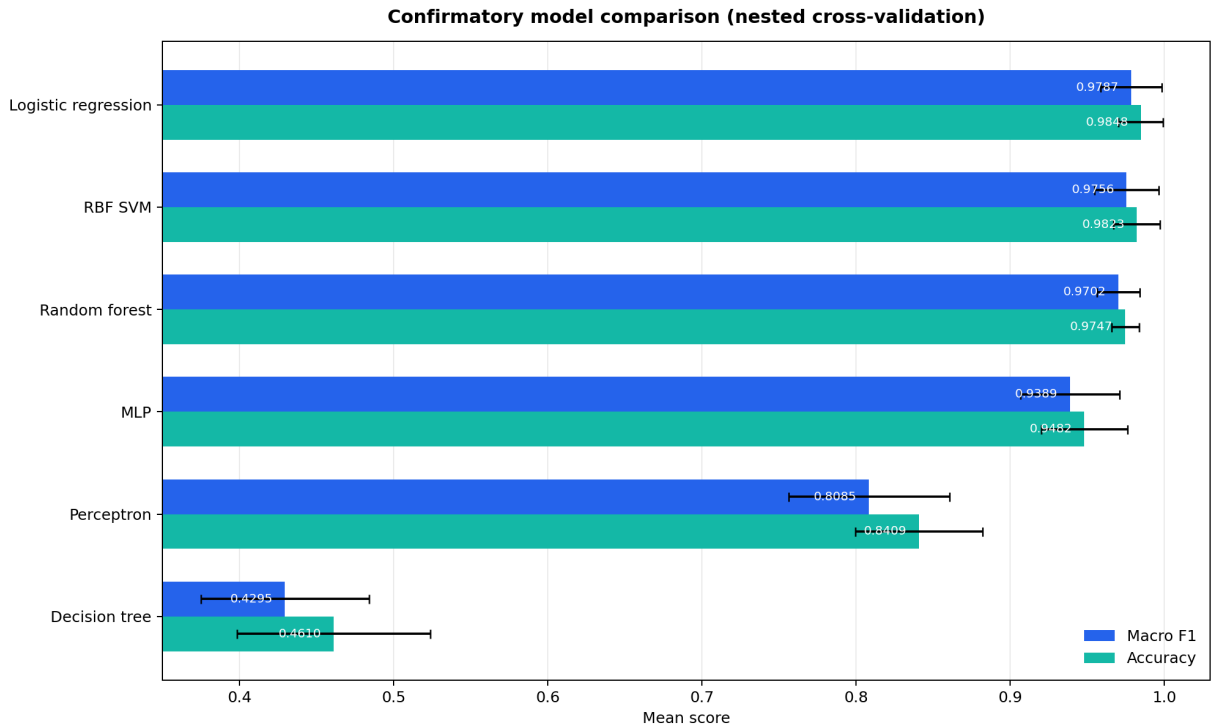


Figure 34: Confirmatory inter-model comparison

A clear hierarchy emerges. Logistic regression and the RBF SVM form the leading confirmatory pair, with a slight average advantage for logistic regression. The random forest also performs at a very high level, although it falls slightly behind. The MLP is a credible solution but does not match the top three models. The perceptron serves effectively as a linear baseline after preprocessing, whereas a single decision tree is clearly inadequate for this multiclass problem.

The key point is the proximity of logistic regression and the SVM. Their mean **macro-F1** scores differ only slightly, although logistic regression retains the higher average. In the absence of a formal statistical test between these closely matched models, it is more prudent to describe them as confirmatory co-leaders, with logistic regression leading on the mean, than to claim a single definitive winner.

4.2 Relative Stability and Practical Trade-offs

Cross-fold standard deviations do not constitute a statistical test, but they provide useful information about the relative stability of the model families under the selected protocol.

Model	F1 range	Stability assessment
Logistic regression	0.0512	Very good stability
RBF SVM	0.0549	Stability comparable to logistic regression
Random forest	0.0343	Most stable of the three leading models
MLP	0.0896	More pronounced variability
Perceptron	0.1121	Insufficient stability for a leading position
Decision tree	0.1449	Very high structural instability

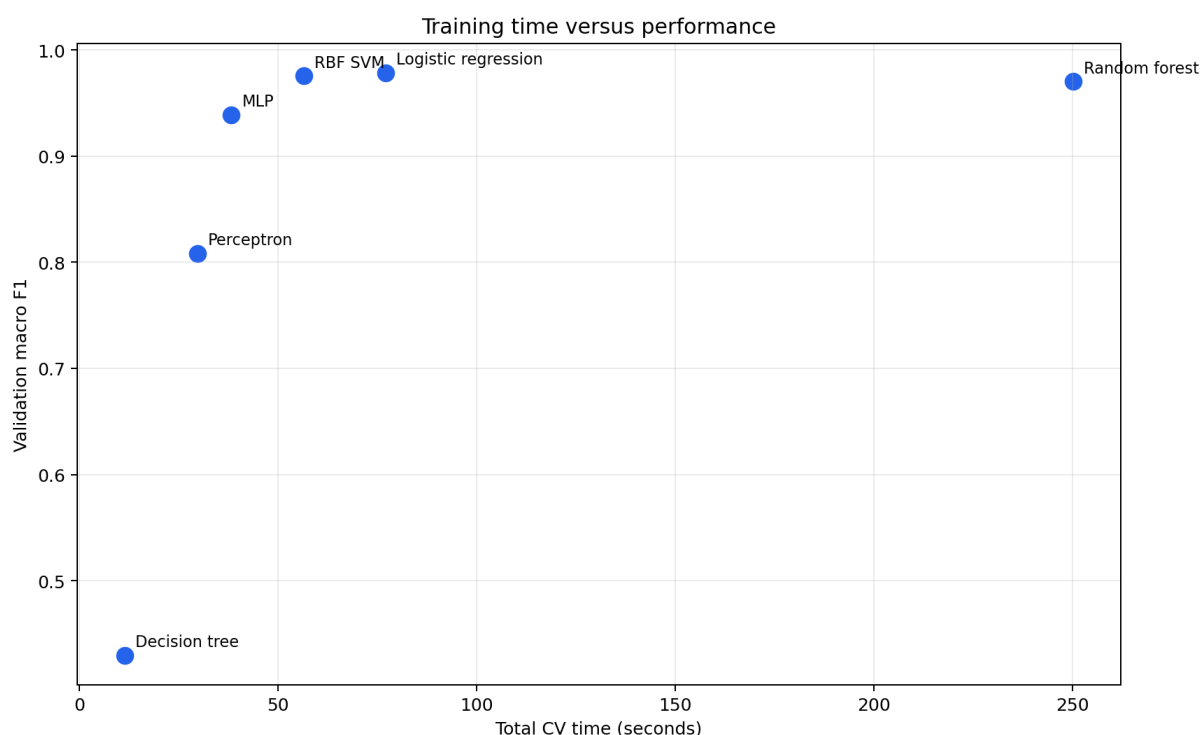


Figure 35: Compute-time/performance trade-off

From a practical standpoint, logistic regression offers the best balance of average performance, computational cost, and interpretability. It achieves the highest observed confirmatory performance while remaining easier to interpret than an RBF SVM and far less expensive than a random forest. The SVM produces a very similar result, but with lower interpretability. The random forest confirms its robustness, although its computational cost is by far the highest among the leading models.

The MLP and perceptron are relatively fast, but that advantage does not offset their performance deficit relative to the strongest candidates. Finally, although the decision tree is inexpensive, its weak predictive performance prevents it from being a defensible solution for this project.

4.3 Cross-Model Error Analysis

The **tuned** OOF predictions on the development subset ($n = 792$) complement the overall analysis with a more detailed view of the number and profile of errors.

Model	Correct predictions	Errors	Error rate
Logistic regression	780	12	1.5 %
RBF SVM	778	14	1.8 %
Random forest	772	20	2.5 %
MLP	751	41	5.2 %
Perceptron	666	126	15.9 %
Decision tree	365	427	53.9 %

This perspective confirms the hierarchy observed for **macro-F1**. The two co-leaders make very few errors across the development set, the random forest remains close behind, and a more substantial gap then appears before the MLP, perceptron, and especially the decision tree. In other words, the difference between the strongest and weaker models is not limited to small score fluctuations: it corresponds to a change in the order of magnitude of the error count.

The overlap among the errors made by the three strongest models provides further insight.

Comparison	Shared errors	Error union	Jaccard
Logistic regression vs. SVM	10	16	0.625
Logistic regression vs. random forest	5	27	0.185
SVM vs. random forest	5	29	0.172

Logistic regression and the SVM therefore share a substantial portion of their residual errors. The overlap is high: 10 shared errors out of 16 distinct errors in total, yielding a Jaccard index of 0.625. This proximity is consistent with their nearly identical performance and suggests that they exploit very similar discriminative structures. The random forest shares fewer errors with either model, supporting the idea that it has a partially different error profile without altering its slight disadvantage on the primary metric.

Overall, the cross-model error analysis suggests that the remaining gains no longer depend on coarse global separation, but on a small number of ambiguous cases. The strongest models therefore differ less in how they represent the general nature of the problem than in their ability to handle these residual decision-boundary regions.

4.4 Secondary Corroboration on the Frozen Set

The frozen corroboration set provides a secondary external check of the selected pipelines. It replaces neither the confirmatory logic of **nested CV** nor the hierarchy established above.

Model	Reference variant	Reference corroboration F1	Tuned corroboration F1	Assessment
Logistic regression	scaled_only	0.9892	0.9892	Neutral tuning effect on corroboration
RBF SVM	scaled_only	0.9838	0.9946	Favorable tuning effect on corroboration
Random forest	default	0.9842	0.9842	Neutral tuning effect on corroboration
MLP	scaled_only	0.9785	0.9785	Neutral tuning effect on corroboration
Perceptron	scaled_only	0.8815	0.8694	Slightly unfavorable tuning effect on corroboration

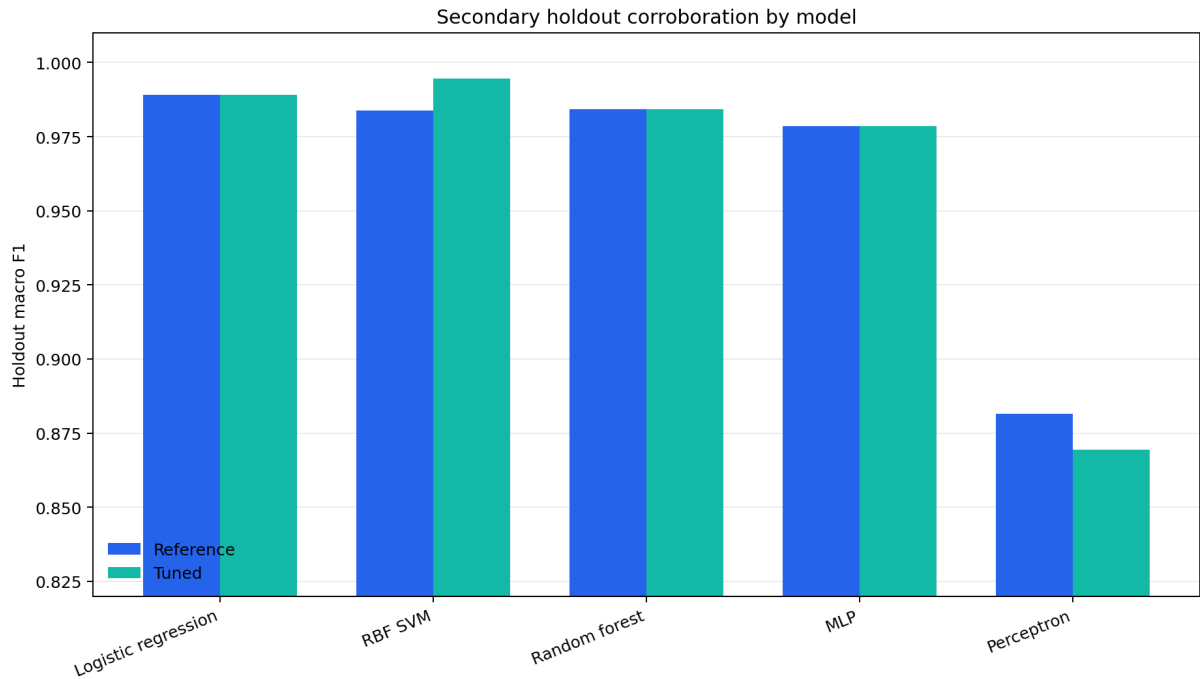


Figure 36: For each eligible model, comparison of the reference variant selected on the development set with the **tuned** variant on the frozen corroboration set.

Considered in isolation, these figures might suggest reranking some models. Such an interpretation would be methodologically unsound. The corroboration set does not represent a new selection stage here, but rather a final secondary check on a single split. It broadly confirms the strength of the leading pipelines, but it should not be used to overturn the primary hierarchy based on confirmatory nested cross-validation estimates.

4.5 Scientific Discussion

Several general lessons emerge from the study. First, a well-regularized linear model can remain extremely competitive in a high-dimensional multiclass problem when paired with suitable preprocessing. Logistic regression shows that a relatively parsimonious structure can be sufficient to achieve a very high level of generalization.

Second, the RBF SVM confirms the presence of exploitable nonlinear structure, but provides no measurable improvement over logistic regression. In this context, practical criteria—particularly interpretability and computational cost—can therefore guide the choice between the two.

Third, the random forest also warrants attention. Its very strong performance without scaling shows that an aggregated tree-based approach can capture the problem structure effectively. However, its substantial additional cost and slight deficit in **macro-F1** make it a less natural choice when the objective is the best overall trade-off.

The MLP results are also instructive. The model becomes competitive after normalization, but it does not outperform the strongest, simpler approaches. This observation is consistent with a dataset that is relatively modest in size given the multiclass complexity of the problem: greater representational capacity does not automatically yield better generalization.

Finally, the contrast between the decision tree and random forest underscores the decisive role of aggregation. The interpretability of a single tree is not enough when the model’s variance becomes excessive. Here, the individual tree primarily serves as a useful counterpoint that illustrates what the ensemble corrects.

In summary, the project demonstrates above all that a well-regularized linear model can remain highly competitive when supported by appropriate preprocessing. The gains from more flexible models are modest here and must be assessed in light of their cost, stability, and interpretability.

5 Conclusion

This study compared six model families on the **Leaf Classification** dataset within a framework that explicitly separates exploration, confirmation, and secondary corroboration. The primary ranking is based on the **tuned** variants evaluated through nested cross-validation, with **macro-F1** as the reference metric.

The results confirm a leading pair consisting of logistic regression and the RBF SVM, with a slight average advantage for logistic regression and the random forest close behind. The MLP remains credible without matching this trio, while the perceptron and especially the decision tree lag behind.

From a practical standpoint, logistic regression is the most defensible choice because it combines the highest observed mean score, moderate computational cost, and greater interpretability than the closest nonlinear alternatives. The frozen corroboration set broadly supports this interpretation without redefining it.

This conclusion is nevertheless context-specific: it is based on a single dataset, without independent external validation, and on one nested cross-validation scheme. The most natural extensions are therefore to repeat the evaluation across multiple data splits, test the robustness of the conclusions on external data, and examine in greater depth the residual errors shared by the strongest models.

6 Appendices

6.1 Purpose and Use of the Appendices

The appendices bring together material that supports reproducibility, methodological auditing, and detailed examination of the artifacts without overloading the main body of the report. They extend the analysis without shifting its inferential basis. The principal conclusions remain grounded in the **tuned** variants evaluated with **nested_cv**; depending on the case, the other information retained here is descriptive, exploratory, or documentary.

6.2 Detailed Protocol and Canonical Artifacts

Element	Selected value
Global seed	42
Frozen split	80 % development / 20 % corroboration
Development set size	792
Corroboration set size	198
Primary cross-validation	5 outer folds
Internal tuning	3 folds
Variance retained by PCA	0.99
Primary metric	Macro-F1
Secondary metric	Accuracy
CV parallelism	CV_N_JOBS = 1

Table 14: Experimental protocol parameters

Preserving a frozen split separate from the development subset maintains a clear distinction between the primary analysis and secondary corroboration. The non-**tuned** variants are used to explore the effects of preprocessing and model capacity; only the **tuned** variants form the basis of the inter-model ranking.

6.3 Detailed Results by Variant

Model	Variant	Status	Validation macro-F1	Validation accuracy	Time (s)
perceptron	default	Exploratory	0.6979	0.7436	0.62
perceptron	scaled_only	Exploratory	0.8228	0.8460	0.60
perceptron	scaled_pca	Exploratory	0.8057	0.8346	0.65
perceptron	tuned	Confirmatory	0.8085	0.8409	29.79
regression_logistique	default	Exploratory	0.2591	0.3106	1.69
regression_logistique	scaled_only	Exploratory	0.9798	0.9861	8.10
regression_logistique	scaled_pca	Exploratory	0.9784	0.9848	2.21
regression_logistique	tuned	Confirmatory	0.9787	0.9848	77.05
svm	default	Exploratory	0.8389	0.8561	2.36
svm	scaled_only	Exploratory	0.9712	0.9785	2.45
svm	scaled_pca	Exploratory	0.9712	0.9785	2.50
svm	tuned	Confirmatory	0.9756	0.9823	56.49
foret_aleatoire	default	Exploratory	0.9764	0.9785	9.16
foret_aleatoire	restricted_depth	Exploratory	0.9345	0.9445	5.42
foret_aleatoire	tuned	Confirmatory	0.9702	0.9747	250.11
arbre_decision	default	Exploratory	0.5809	0.6174	1.04
arbre_decision	restricted_depth	Exploratory	0.1468	0.1529	0.56
arbre_decision	tuned	Confirmatory	0.4295	0.4610	11.47
mlp	default	Exploratory	0.1642	0.1863	1.97
mlp	scaled_only	Exploratory	0.9389	0.9482	2.01
mlp	scaled_pca	Exploratory	0.9323	0.9444	1.97
mlp	tuned	Confirmatory	0.9389	0.9482	38.25

Table 15: Validation results by variant

This table must be read with due regard for the status of each row. Scores for the exploratory variants document the effects of preprocessing and complexity; they must not be conflated with the confirmatory scores of the **tuned** variants.

6.4 Hyperparameters Selected by Outer Fold

The stability of fold-level selection provides useful evidence about the robustness of the model families. Some families converge on the same settings in nearly every fold, while others show more pronounced variability, particularly in the use of **PCA** or the strength of regularization.

Model	Patterns observed across the 5 outer folds
perceptron	<code>alpha = 0.001</code> , <code>penalty = l1</code> in every fold; PCA selected once, otherwise <code>passthrough</code>
regression_logistique	C alternates between 1.0 and 10.0; PCA selected in one fold, otherwise <code>passthrough</code>
svm	C = 10.0 and <code>gamma = scale</code> in every fold; PCA selected once
foret_aleatoire	Nearly perfectly stable settings: <code>max_depth = None</code> , <code>max_features = sqrt</code> , <code>min_samples_leaf = 1</code> , and <code>n_estimators = 400</code> in every fold
arbre_decision	<code>max_depth = 20</code> , <code>criterion = gini</code> , and <code>min_samples_leaf = 1</code> ; only <code>ccp_alpha</code> varies, taking 0.005 in one fold and 0.0 otherwise
mlp	<code>hidden_layer_sizes = [128]</code> in every fold; <code>alpha = 0.0001</code> in four folds and 0.001 in one fold; PCA is never selected

Table 16: Hyperparameters selected by outer fold

The random forest, SVM, and MLP show highly consistent fold-level selection. Logistic regression and the perceptron remain more sensitive to linear regularization and the possible selection of PCA, but this variability does not alter their overall interpretation. For the decision tree, stability in the selected depth does not offset the predictive instability observed in its scores.

6.5 AI Use Declaration

This declaration is provided for documentary transparency. AI assistance was used on a limited basis during the project, but it did not replace scientific decision-making, the execution of experiments, or the final validation of results.

Where AI was used

- assistance with structuring Markdown and LaTeX documents;
- occasional assistance with repository cleanup, reorganization, and documentation;
- technical assistance with certain adjustments to code, scripts, and notebooks;
- proposals or initial generation of some code sections, which were subsequently reviewed, modified, tested, and validated by a human before being retained.

Work performed manually

- definition of the topic, objectives, and experimental protocol;
- selection of the models, variants, metrics, and evaluation framework;

- execution of the experiments, notebooks, and pipeline;
- review, correction, and validation of the retained code;
- verification of the numerical results, figures, and their consistency;
- selection of the material included in the final report;
- scientific interpretation of the results and final writing validated by a human.

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